

Expert Systems with Applications

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Exploiting Social Capital for Improving Personalized Recommendations in Online Social Networks

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Abstract

This article presents a novel approach to enhancing personalized recommendations in online social networks by exploiting the concept of Social Capital. Recognizing the challenges posed by information overload on platforms like Twitter, the proposed method integrates user interactions and features on social media with the concept of Social Capital to generate more relevant recommendations. The model incorporates a multifaceted analysis of user data, including the user's reputation, influence, and engagement strength, alongside tweets' recency, diversity, and context scores, to calculate a comprehensive Social Capital Score for each recommendation. An extensive offline evaluation of the model demonstrates its effectiveness in improving the personalization and relevance of recommendations. The findings highlight the significant role of incorporating Social Capital in recommender systems, contributing to a more engaging and informative user experience on social media platforms. This study provides valuable insights into the potential benefits and limitations of this approach, advancing the field of recommender systems and enhancing our understanding of the role of social capital in shaping user behavior and preferences.

Keywords: Recommender Systems, Social Capital, Social Networks

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1. Introduction

Social Capital has a long history of studies highlighting how others influence people. The most prominent studies are in philosophy area approaching benefits and advantages on human life such as reciprocity and influence (Putnam, 2000), trustworthiness (Coleman, 1990), access of information (Bourdieu, 2002), and interactions (Lin, 2001). They are useful indicators for describing individual preferences, information’s power, and how people make decisions in real life.

Nowadays, the exponential growth of Web 2.0 has brought those aspects into Online Social Networks (OSN), providing a virtual environment for users to share information, opinions, interests, and make connections with others (Kapoor et al., 2017). For instance, on microblogging Twitter, more than 500 million tweets are sent by 229 million active users daily, producing a high volume of information consumed by other million users.

Even though OSNs provide a wide environment to communicate and relate with others, we can notice the growth of the published content over the long years, overloading users most of the time. Such energy is generated by social interactions and accessible resources, which have been studied in the literature as Social Capital (SC). This concept has been formalized by Bourdieu (2002) as an aggregate value of the resources available within a network of relationships. In addition, Nahapiet & Ghoshal (1998) argue that social network features can be analyzed to model user preferences. Social Capital assumes that individuals benefit from the various norms and values that a social network promotes and produces, including trust, reciprocity, information, and cooperation (Spratt et al., 2018; Coleman, 1990). The accumulation of SC through social network interactions and relationships can provide valuable information about a user’s interests and preferences.

However, sometimes users follow others for many reasons, not only to consume everything they post. For example, to follow friends or family members who present distinct interests and can publish irrelevant content often. In other words, the feeling is that users do not want to unfollow them. They want to

receive personalized content according to their interests. All these observations address one of the most challenging problems in online social networks: Information Overload, which is the natural human incapacity to process the huge amount of information produced in social networks (Zhang et al., 2020). Thus, it is necessary to identify and filter these posts to suggest the most relevant content for an individual.

Recommendation Systems (RSs) have emerged as a solution for the information overload problem by presenting relevant content based on their preferences (Ricci et al., 2015). The main motivation behind building these systems is personalization (Ge et al., 2019). However, traditional Recommendation System (RS) using content-based or collaborative filtering techniques have limitations. Personalized recommendations may be insufficient, and the recommended content may lack diversity, resulting in popular news that has already been widely shared. Additionally, biased recommendations may promote highly shared news within a particular social group, creating a “filter bubble” effect and reinforcing biases (Lunardi et al., 2020). On the other hand, SC can improve the accuracy of RSs by recommending content that the user is likely to enjoy based on their social connections and interactions, besides improving the diversity of recommendations. Moreover, it can help to overcome the cold start problem, where there is often little or no information about new users’ preferences, making it difficult to provide relevant recommendations (Gonzalez-Camacho et al., 2022). Recent studies indicate that integrating information about users’ relationships can lead to more precise and diverse recommendations by providing valuable insights into their interests, preferences, and expertise (Guo et al., 2022; Singh et al., 2022).

Besides relationships, by identifying the most influential users in a network, recommendations can be targeted to users who are more likely to be interested in a particular item, and thus increase the chances of it being recommended to the right users (Zhuang et al., 2021). An example of this influence is when users have many followers or a high frequency of interactions in their posts, affecting other users by their actions inside a network (Morone & Makse, 2015).

For instance, Elon Musk who is an influencer with more than 108M followers, added on his Twitter bio *#bitcoin*, the cryptocurrency spiked 20% after that at the time¹.

This paper exploits factors that can be used to improve the quality of recommendations and aims to investigate, model and account for interactions and available resources on social networks through exploiting social capital. This approach aims to analyze and boost news with more interactions, comments, and influential users across a set of news, which can be interesting for a user to suggest the most relevant.

The main contributions of this work are:

- **The social capital model**, which can help improve recommender systems' personalization. By taking into account a user's social connections and social activities;
- **The user model** that comprises a user profile from user interactions and other features;
- **The offline evaluation**, which can provide valuable insights into the potential benefits and limitations of this approach and help to advance the field of recommender systems by incorporating insights from social network analysis and enhancing our understanding of the role of social capital in shaping user behavior and preferences.

This paper is structured as follows: Section 2.2 presents background on Recommendation Systems and Social Capital. Section 3 discusses related works. Section 4 introduces the proposed recommendation algorithm. Section 5 presents the evaluation of the application, as well as the experimental setup and results. Finally, Section 6 concludes the work and presents future works.

¹<https://www.cnn.com/2021/01/29/bitcoin-spikes-20percent-after-elon-musk-adds-bitcoin-to-his-twitter-bio.html>

2. Background

This section provides an overview of two important topics in the field of social networks: Recommender Systems and Social Capital.

2.1. *Recommender System*

In recent years, RSs have become a critical tool for enhancing user engagement and satisfaction for social networks. These systems use algorithms and techniques to analyze users' interactions and behaviors to generate personalized recommendations. By understanding users' preferences, RSs can provide more relevant and accurate content, thus increasing user satisfaction and retention.

One of the key aspects of RSs in social networks is the analysis of user interactions. Social networks generate vast amounts of data, including user behavior, preferences, and interests. By analyzing this data, it is possible to identify patterns and relationships between users and items, which can be used to generate personalized recommendations (Ricci et al., 2015). For example, Adomavicius & Tuzhilin (2005) describe collaborative filtering as an approach that uses user-item interactions to generate recommendations. In addition to analyzing user interactions, RSs can also contribute to creating user profiles. User profiles represent users' interests, preferences, and behaviors. By analyzing user interactions, the system can create a detailed user profile that can be used to generate more accurate recommendations. For instance, He et al. (2017) propose a neural collaborative filtering approach that uses user-item interactions to learn latent user and item features, which can be used to generate personalized recommendations.

Furthermore, RSs can also help identify new and emerging trends in social networks by analyzing user interactions and behaviors. This information can improve the accuracy and relevance of recommendations and provide insights into user behavior and preferences. Bobadilla et al. (2013) describe how content-based filtering can identify new items that match users' preferences, even if they have not interacted with them.

Recommender systems use various techniques to provide personalized recommendations to users based on their preferences and past behavior. In addition, several state-of-the-art techniques in the literature, such as collaborative and content-based filtering, are the two most commonly used approaches in RSs (Adomavicius & Tuzhilin, 2005; Bobadilla et al., 2013). However, recent studies have proposed several other techniques, including matrix factorization, deep learning, and hybrid approaches, that combine multiple techniques to improve the accuracy and coverage of recommendations.

Hybrid approaches that combine multiple techniques, such as collaborative and content-based filtering, have also been proposed to improve the accuracy and coverage of recommendations (Adomavicius & Tuzhilin, 2015). For example, a hybrid system could use content-based filtering to recommend items to new users or users with limited interaction history and collaborative filtering to recommend items to more active users with more substantial interaction history.

Although several techniques have been proposed to improve the accuracy and coverage of recommendations, there is still a gap in research regarding incorporating social context and network structure in RSs. Social network structures and the influence of users on each other can significantly impact the recommendations provided by these systems. Therefore, by incorporating the social network features, the Social Capital model could lead to more effective recommendations. Moreover, these systems can help build user trust and understanding, increasing user satisfaction and engagement.

2.2. Social Capital

The role of social capital in RSs has been widely studied in recent years, focusing on understanding how social connections and relationships can be leveraged to improve the quality of recommendations. Social capital, which refers to the resources individuals can access through their social connections and relationships, is a key factor in the effectiveness of recommendation systems (Lin, 2001). Recommender systems that use SC can make recommendations based on the connections and interactions of users within a social network. For example,

a tweet recommendation system using SC might recommend tweets to a user based on the tweets their friends and followers liked or shared. In this way, the system can leverage the SC of the user’s network to make more accurate and relevant recommendations.

The benefits and advantages of relationships or interactions with publications on social networks, such as access to information, are defined by Lin (2001) as Social Capital. According to Bourdieu (2002), Social Capital can be defined as “the sum of accumulated resources through the relationships among users within in a network”. In this case, we can notice that SC has an embracing concept, allowing us to use various approaches to analyze and identify ways to calculate news power on social networks.

Another meaningful resource is social influence (Rezaie et al., 2020), which is an important factor in recommender systems, as it can provide valuable information about the preferences and relationships of users. Recommender systems that consider users’ social influence can make more accurate and relevant recommendations as they consider different users’ influence on one another. Several approaches can be used to incorporate social influence into a recommender system. One common approach is using graph-based (Chen et al., 2022) methods to model the structure of the social network and the influence different users have on one another. These methods can be used to identify key influencers within the network and to make recommendations based on these users’ influence on their followers. We use the user’s influence score to calculate news value in our work.

According to Li & Xiong (2017), the number of mentions and comments in the news can contribute to measuring the influence among users in a social network. Furthermore, the total of the user’s followers can also contribute to identifying his influence in the entire social network. In this case, they developed a Recommender System that uses direct interactions among users as a recommendation method to analyze the strength of the connections among them to measure their preferences and the item’s characteristics to be recommended. In this way, they use two approaches to measure the user’s influence: the macro-

scopic and microscopic social influence, which are used in a matrix to apply matrix factorization techniques to generate more precise recommendations.

Finally, another approach to be used as a SC resource is to use Natural Language Processing (NLP) techniques to analyze the content of news and other social media posts to identify users’ preferences and interests. This information can be used to make recommendations based on users’ interests and influence on their followers (Girgin, 2021).

3. Related Works

In the realm of Recommender Systems (RSs), various methodologies have been explored, each with its unique approach to enhancing personalized recommendations. The study by He et al. (2017) exemplifies this with their neural collaborative filtering method, which utilizes user-item interactions to discern latent user and item features for personalized recommendation generation. This contrasts with our study’s approach, which focuses on integrating social capital elements like user influence and engagement, thus providing a recommendation system that is not just based on item-user interactions but also enriched with social context and user behavior dynamics.

A recent study by Cui et al. (2023) explored the application of a gated recurrent neural network for enhancing recommendation systems. They focused on constructing a user preference model that integrates the Preference Attention Model Based on Social Relations (PASR). This approach aims to extract user social influence preferences, achieving a nuanced blend of user preference and social influence through their Recommendation Algorithm Based on User Preference and Social Influence (UPSI). This study shares similarities with our research in its focus on utilizing social influence and user preferences to improve recommendation systems. However, our research distinguishes itself by its comprehensive integration of various social capital elements, such as user reputation and engagement, alongside influence. This multi-faceted approach in our approach offers a broader and more detailed understanding of user be-

havior and preferences, potentially providing a more holistic view compared to the singular focus on preference and influence fusion in the UPSI algorithm. Bakshy et al. (2012) explore how social networks aid in the spread of information, focusing particularly on the influence of users in disseminating information. This contrasts with our study, which considers user influence and integrates a broader range of social capital elements for personalized recommendations. Our approach, therefore, provides a more nuanced and user-centric model that goes beyond information diffusion to encompass various aspects of social interactions.

Traditional methods in RSs, as described by Bobadilla et al. (2013) and others, include content-based and collaborative filtering, along with hybrid approaches that amalgamate multiple techniques. These conventional methods, while effective, primarily focus on user preferences and past behaviors. Our approach, however, extends these traditional methods by integrating aspects of social capital, such as user influence and the structure of social networks. This integration offers a more nuanced recommendation model that caters to user preferences and takes into account the user’s social environment, providing a more comprehensive and user-centric recommendation experience. The role of social capital in RSs has been a focal point in various studies, with researchers like Lin (2001), Bourdieu (2002), and Rezaie et al. (2020) emphasizing the significance of leveraging social connections and relationships to enhance recommendation quality. These studies underscore the importance of social capital in recommendation systems, a perspective our study aligns with. However, our approach uniquely quantifies social capital through specific metrics such as influence scores and combines it with social network analysis and natural language processing techniques. This results in a more precise and socially aware recommendation system beyond leveraging social connections.

Researches like those by Chen et al. (2022) have proposed graph-based methods to model the structure of social networks and the influence of users within these networks. These methods are critical in shaping the nature of the recommendations made by the systems. Our proposal parallels this approach in considering user influence, but it goes further by integrating it into a broader

framework of social capital metrics. This includes elements like user reputation and engagement, thereby focusing on influence and considering the overall social dynamics within the network. This comprehensive approach results in a more holistic recommendation model sensitive to the nuances of social interactions and user influence. Lastly, the focus on diversity exposure and user influence, as seen in studies by Tsai et al. (2020) and others, aims at enhancing social capital by diversifying social connections and examining user influence for recommendation purposes. While these studies emphasize the importance of user influence in recommendation systems, our study broadens this scope. It combines user influence with other elements of social capital, like reputation and engagement, to create a model that recommends based on user preferences and the intricate social dynamics within the network. This results in a more contextually rich and socially informed recommendation system, distinguishing our study from others in the field.

However, thus far, the exploitation of SC on social networks to improve recommendations quality has been neglected. In this case, this work aims to analyze and find resources on Twitter that can be used to build a recommendation model and generate relevant recommendations according to user’s preferences. Consider that this research can be extended to any social network that applies relationships and interactions.

4. The Proposal

This section outlines the proposed RS by this paper. Firstly, the recommendation overview is presented. After that, the fundamental features of Twitter are presented. Following this, the news and users are modeled, and the discussion moves to the user’s influence and reputation. Finally, is presented the recommendation model algorithms.

4.1. Recommendation Overview

The proposed recommender system aims to generate recommendations based on Twitter data. In addition, it uses a variety of metrics to assess the social

capital of Twitter users and their tweets, including the user’s influence score, reputation score, recency score, and context score. These scores are calculated based on metrics such as the user’s follower count, the number of tweets and their average engagement rate, the sentiment of tweets mentioning or replying to the user, and the relevance of the tweet’s content to the topic. It also considers the diversity and uniqueness of the tweet’s content and the presence of relevant media, such as images or videos. By assessing these various factors, the algorithm generates a social capital score for each tweet, allowing it to recommend those likely to be more relevant to users (with the highest score). Figure 1 presents an overview of the recommendation system proposed by this paper.

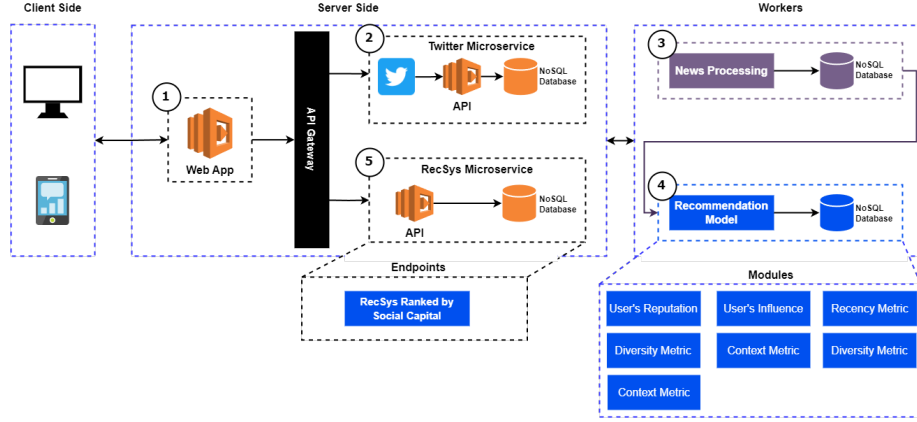


Figure 1: Steps to generate personalized recommendations.

1. **Web App:** This step comprises a web app, which provides the business logic to auth and generate recommendations for each user;
2. **Twitter Microservice:** This service is responsible for connecting the user’s Twitter account with the recommender system to analyze their data;
3. **News Processing:** This background service is responsible for analyzing and pre-processing all extracted news;
4. **Recommendation Model:** In this step, all recommendation metrics were calculated for each pre-processed news;

5. **RecSys Microservice:** This is the service responsible for getting the recommendations for each authenticated user.

4.1.1. Motivation Scenario

This section showcases the application of the proposed RS. The idea is to present a practical example of the algorithm’s utility in analyzing and evaluating SC metrics for two simulated tweets. The analysis will walk through the entire recommendation process, including all the metrics used, such as the author’s followers, likes, replies, impressions, retweets, mentions, media, userStrength-Score, and more. This analysis is intended to demonstrate how the algorithm can be applied in practice in assessing SC in various contexts. Figure 2 presents an example of two tweets and the user profile.



Figure 2: Motivation Scenario.

- **Tweet A:** *Excited to share our latest research on the role of social capital in improving recommender systems! Our findings show that incorporating user social networks can significantly improve recommendation quality. #research `http: \url01 http: \url02 http: \url03`;*
- **Tweet B:** *Just read an interesting article about the importance of social capital in recommender systems. It's amazing how much impact a user's*

social connections can have on the quality of recommendations they receive!

#socialcapital #recommendations http: \url01 http: \url02.

Several metrics were considered in a scenario analyzing the motivations behind two simulated tweets. For Tweet A, there were 10 likes, 1 retweet, no replies, 79 impressions, 1 hashtag, 3 URLs, 3 media, a diversity score of 0.98, a user author strength of 0, a mention users strength of 0.53, 40 tokens in length, a context score of 0.28, and a recency score of 0.66, resulting in a social capital of **92.04**. On the other hand, Tweet B received 1 like no retweets or replies, 119 impressions, 2 hashtags, 2 URLs, 1 media, a diversity score of 0.98, a user author strength of 4.46, no mention users strength, 42 tokens in length, a context score of 0.29, and a recency score of 0.77, resulting in a higher social capital of **132.62**.

Based on these scenarios, we can see that **Tweet B** has a higher social capital score than Tweet A, indicating that it is likely to be more influential and have a greater impact within the social network.

4.2. Features of Twitter

Twitter is a social media platform that allows users to send and read short messages called “tweets”. Tweets are limited to 280 characters and can include text, images, and videos. Twitter was founded in 2006 and has become a popular tool for sharing information, news, and opinions. It has over 330 million monthly active users and is available in over 40 languages. Users can follow other users, receive updates from them, and engage in public conversations by replying to and retweeting tweets. Twitter is also often used as a tool for marketing, customer service, and activism.

Tweets: The core feature of Twitter is the ability to create and post tweets. Tweets are limited to 280 characters and can include text, images, and videos;

Retweets: Users can retweet tweets from other users to share them with their followers. This is a way to amplify the reach of a tweet and increase its visibility;

Likes: Users can like tweets to indicate their approval or support for the content;

Replies: Users can reply to tweets to engage in conversation with other users; **Hashtags:** Hashtags are keywords or phrases preceded by the “#” symbol, which can be used to categorize and search for tweets related to a particular topic; **Mentions:** Users can mention other users in their tweets by including the “@” symbol followed by their username. This is a way to notify the other user and potentially start a conversation; **Lists:** Users can create lists of other Twitter users to group them by topic or interest. This is a way to see tweets from a specific group of users easily.

4.2.1. Twitter’s API

Twitter offers an API that enables users to access information on its platform. This API provides endpoints that allow users to extract the content of a timeline, post news, interact with others, and follow any account. Twitter also offers access plans with certain limitations on data extraction over a given period, denoted as t . For our study, we utilized the academic access plan, allowing up to 450 requests per 15-minute window.

4.3. Notation

This section aims to provide a quick reference to understand better the meaning of the used variables and their role in the proposed methods. Understanding the notation is crucial for comprehending the mathematical equations and algorithms presented in the text. Specifically, we use the following notations presented in Table 4.3.

In a social network RS, both the user and news model can be defined by a set of features and metadata that describe each user’s behavior, interests, and preferences. These features can be extracted from the user’s profile and interactions with the platform, including the content they create and the accounts they follow and engage with. In addition, the tweet’s content, along with various metrics related to its engagement and relevance to the user, should be considered. The recommender system can provide more personalized and relevant

Notation	Description
u	Represents a Twitter user
$t_m(u)$	Represents a list of user u 's most recent mentions
$list(u)$	Represents the number of public lists the user is a member of
$followers(u)$	Denotes the number of followers of a user
$t(u)$	Represents the list of a user's most recent posts
$t_r(u)$	Represents a list of user u 's most recent replies
$s_{m,i}$	Represents the text of the i -th mention in $t_m(u)$
$s_{r,j}$	Represents the text of the j -th reply in $t_r(u)$
$f_m(s)$	Represents the sentiment score of text s
l	Represents the number of likes
r	Represents the number of retweets
q	Represents the number of quotes
p	Represents the number of replies
im	Represents the number of impressions

Table 1: Notations, which are used in the next sections.

recommendations by analyzing these different factors. In this case, Users and News can be represented as:

1. **Users (U)**: Social networks are constituted of interlinked users that post, comment, mention, or forward some news. Users are represented as $U = \{u_1, u_2, \dots, u_n | 1 \leq n \leq \mathbb{N}\}$, where each $u \in U$ is defined as a tuple $u = \langle t, t_m, t_r, s_t, s_r, s_m, f_m(s), list, l, r, q, p \rangle$ with twelve metadata, which are used to calculate the reputation and influence of users (see Section 4.5);
2. **News (T)**: News are the recommended items for a given user, and they are represented as $T = \{t_1, t_2, \dots, t_z | 1 \leq z \leq \mathbb{T}\}$, where each $t \in T$ is a tuple $t = \langle s, l, r, q, p, im, urls, emojis, media, hashtags, created_at \rangle$ with eleven metadata, which are used for calculating the social capital score (see Section 4.6);

Algorithm 1 Computation of Enhanced Social Capital Score

Require: df_{result} , DataFrame with tweet metrics

Ensure: df_{result} updated with Social Capital Scores

$df_{\text{result}}['\text{normalized_sentiment}'] \leftarrow \text{scaler.fit_transform}(df_{\text{result}}[['\text{sentiment_score}']])$

$df_{\text{result}}['\text{engagement_score}'] \leftarrow \text{mean}(df_{\text{result}}[['\text{like_count}', '\text{retweet_count}', '\text{reply_count}', '\text{quote_count}']], \text{axis}=1)$

Calculate mentions_count and urls_count for Network Influence (NI)

$df_{\text{result}}['\text{content_virality}'] \leftarrow df_{\text{result}}['\text{retweet_count}'] + df_{\text{result}}['\text{quote_count}']$

Define weights: $\alpha, \beta, \gamma, \delta, \epsilon, \zeta \leftarrow 0.20$

Compute SCS_{new} using defined weights and component scores

Normalize SCS_{new} across df_{result}

return df_{result}

4.4. Data Modeling

In this section, we describe the different techniques and algorithms used to pre-process news and generate meaningful features for modeling, besides how the user profile is built.

4.4.1. News Pre-Processing and Modeling

These steps are crucial to cleaning and standardizing the raw text, reducing dimensionality, and extracting the relevant information needed for the models to perform well. Figure 3 shows an overview of the techniques used by this step.

1. Cleaning:

- **Convert text to lowercase:** This step is important because it ensures consistency when comparing or analyzing words in the text. For example, “recommendation” and “RECOMMENDATION” would be treated as different words if not converted to lowercase;
- **Remove URLs:** Removing URLs is important because they do not provide useful information in analyzing the tweet’s content. For ex-

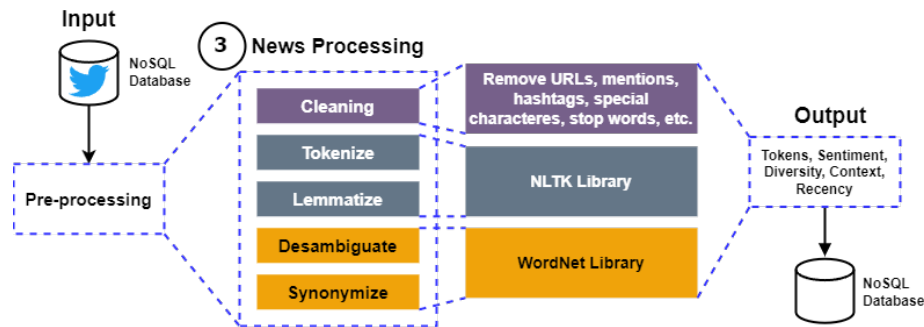


Figure 3: Steps to pre-process news.

ample, “Check out this website: example.com” would be reduced to “Check out this website.”;

- **Remove mentions and hashtags:** Removing mentions and hashtags is important because they are not relevant to the content of the tweet itself. For example, “Great article by @user1 on #AI” would be reduced to “Great article by on”;
- **Remove punctuation:** Removing punctuation is important because it can interfere with the analysis of the text. For example, “Hello, world!” would be reduced to “Hello world”.

2. **Tokenize:** Tokenization breaks down the text into individual words or tokens. This is an important step because it analyzes individual words within the text. For example, “This is a tweet” would be tokenized into [“This”, “is”, “a”, “tweet”];
3. **Remove stop words:** Stop words are words commonly used in a language but do not add significant meaning to the text. Removing stop words is important because it reduces the noise in the text and helps focus on the most important words. For example, “I am a big fan of pizza” would be reduced to [“big”, “fan”, “pizza”];
4. **Lemmatize:** Lemmatization reduces words to their base form or lemma. This is important because it reduces the number of unique words in the

text and helps group related words together. For example, “running”, “ran”, and “runs” would all be reduced to “run”;

5. **Disambiguate polysemous words:** Polysemous words are words that have multiple meanings. Disambiguation is the process of determining the correct meaning of a word based on the context in which it is used. This is important because it ensures that the correct meaning is used in the text’s analysis. For example, “I need to park my car in the park” could mean either a public area or to leave a vehicle in a particular location;
6. **Expand synonyms:** Expanding synonyms is the process of adding synonyms of words to the text. This is important because it helps capture the text’s full meaning and can help group related words together. For example, “big” could include “large”, “huge”, and “massive”.

4.4.2. User’s Profile Modeling

The user’s profile is a critical component of any recommendation system. It is a representation of the user that includes various features that are relevant for making recommendations. In our recommendation system, the user’s profile is built using a combination of the following features:

$$MU_u = \{features, (word, frequency) | words \in BW, features \in n\} \quad (1)$$

where each $feature_i$ represents a metric of a news item (recency, likes, followers, for example), $word_i$ is a term from a specific news item, and $frequency$ is the value of the term’s relevance obtained through the TF-IDF metric, and BW is the bag-of-words.

4.5. User’s Engagement Strength

The engagement strength of a user on social media platforms is a critical factor in determining their impact and influence within a community. This section proposes methods for measuring user engagement strength on Twitter, a widely used social media platform for communication and information sharing. In this

case, we introduce methods for calculating user reputation and influence scores, considering factors such as sentiment analysis of messages, post frequency, and feedback from followers. The user’s engagement strength is defined by the Equation 2:

$$userStrengthScore(u) = reputationScore(u) \cdot influenceScore(u) \quad (2)$$

where u can be an author, a mention, or a user in a reply, $reputationScore(u)$ and $influenceScore(u)$ are discussed in the following sections.

4.5.1. User’s Reputation

One way to calculate the reputation score of Twitter users is to use sentiment analysis of their recent mentions and replies. The idea behind this method is that the sentiment and tone of messages a user receives or generates on Twitter can reflect their engagement, influence, and reputation.

To calculate a user’s reputation score, we first retrieve their most recent mentions and replies. We then apply sentiment analysis to the text of these messages, which assigns a sentiment score between -1 (negative sentiment) and 1 (positive sentiment) to each message. The sentiment score reflects the overall sentiment or emotion conveyed by the text. We compute the user’s reputation score as the ratio of the sum of the positive sentiment scores to the total number of sentiment scores of the messages. The intuition behind this ratio is that a user with a higher ratio of positive to negative sentiment scores will likely have a better reputation and more positive influence on the platform than a user with a lower ratio. The reputation score is shown by Equation 3:

$$reputationScore(u) = \frac{PM(u) + PR(u)}{PM(u) + NM(u) + PR(u) + NR(u)} \quad (3)$$

where $PM(u)$ is the number of positive sentiment mentions about user u , $NM(u)$ is the number of negative sentiment mentions about user u , $PR(u)$ is the number of positive sentiment replies to user u , and $NR(u)$ is the number of negative sentiment replies to user u . After calculating the reputation

score, a fine adjustment parameter θ is used to adjust the impact of the user’s listed count on the reputation score. The normalized reputation score is then calculated by Equation 4 as:

$$normalizedReputationScore(u) = \frac{reputationScore(u) + \theta \cdot list(u)}{1 + \theta \cdot list(u)} \quad (4)$$

where $list(u)$ is the number of public lists the user is a member of, and θ is a fine adjustment parameter.

The parameter θ modulates the effect of the number of public lists a user is a member of on their normalized reputation score. In this context, the number of public lists a user is a member of is being used as a proxy for their level of influence or prestige on the platform, which is considered alongside their sentiment-based reputation score.

When θ is zero, the normalized reputation score equals the reputation score. In other words, the number of public lists the user is a member of does not affect their normalized reputation score. When θ is increased, the impact of the user’s listed count on the reputation score is increased. A higher value means that being a member of more public lists will significantly and positively affect the normalized reputation score. The setting of θ in experimental analysis or practical applications would be based on how important the number of lists a user is on is deemed for their reputation. If it is considered very important, it would be set higher to reflect this. If it is considered less critical, it should be set lower.

In practice, θ could be set empirically by analyzing how well the normalized reputation score correlates with other measures of user influence and engagement when it is varied. It’s also possible to use machine learning to optimize it so that the normalized reputation score predicts some outcome of interest, such as the likelihood of a user’s content being retweeted or the user gaining followers.

In summary, it is a tuning parameter that adjusts the contribution of a user’s list membership count to their overall reputation score. It should be set based on the relative importance of list membership to reputation within the specific

context of the application.

4.5.2. User's Influence

We propose a new influence metric that combines a user's popularity and activity on the social media platform in terms of post frequency and feedback received from followers. It is calculated using the following Equation 5:

$$influenceScore(u) = \log_{10}(followers(u) + 1) \cdot (engagement(u) + 1) \quad (5)$$

where u is a user, $followers(u)$ is the number of followers of the user, $t(u)$ is a list of the user's most recent posts on the platform, and $engagementScore(u)$ is the average engagement rate of the user. The engagement rate is calculated as shown in Equation 6:

$$engagementScore(u) = \begin{cases} \frac{\sum_{i=1}^{t(u)} (l_i + r_i + q_i + p_i + im_i)}{t(u) \cdot followers(u)} & \text{if } t(u) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where l_i , r_i , q_i , p_i , and im_i are the engagement metrics (e.g., like, retweet, quote, reply, and impressions) of the i -th post in $t(u)$, which are non-negative integers. The influence score is a non-negative real value that measures the user's potential impact, reach, or authority on the social media platform based on their number of followers and engagement rate.

4.6. The Social Capital Metric

We introduce a novel social capital metric that considers several user and content-related features to measure social capital on Twitter. The metric includes the recency score, diversity score, and context score of a tweet and several other features such as the number of likes, retweets, replies, impression count, number of hashtags and URLs, and other tweet features.

4.6.1. Recency Score

The recency score of a tweet can be calculated using a logarithmic decay function that considers the tweet's age. The decay factor determines how quickly

the score decays as the tweet ages, with a higher decay factor resulting in a faster decay rate. The recency score is calculated using the following Equation 7:

$$recencyScore(t) = \frac{1}{1 + a \cdot \log_{10}(1 + t_{age} - dt)} \quad (7)$$

where t is a tweet, a is the decay factor, age is the age of the tweet in seconds, and dt is the current date and time. This equation calculates the age of a tweet with a logarithmic decay that slows down over time. **Take into consideration that the value of a was chosen empirically varying between 0.1-1.0 and 0.1 was adopted in this work.**

The resulting value is between 0 and 1, where 1 represents a very recent tweet, and 0 is a very old tweet. The idea in applying this metric is that it can be better suited in cases where recent tweets are more important than older ones and where a more gradual score decay over time is desirable.

The logarithmic decay function was chosen for the recency score to ensure a more gradual decay over time, prioritizing recent information. This decision was based on the nature of social media dynamics, where recent activities are often more relevant than older ones. The selection of a logarithmic decay function to determine the recency score of a tweet is motivated by several factors:

- **Gradual Decay Over Time:** Unlike linear or exponential decays, a logarithmic decay ensures that the decay rate is faster initially and slows down over time. This mirrors real-world scenarios in many social media applications where the relevance of content is most significant shortly after it's posted and gradually diminishes as it ages. However, even as it ages, it doesn't immediately become completely irrelevant. The logarithmic decay captures this behavior naturally;
- **Flexibility with Decay Factor:** We can control the decay rate by incorporating a decay factor θ into the formula. A higher value of θ leads to a faster decay, allowing for adaptability based on the context in which the metric is being applied;

- **Natural Bounding:** The resultant recency score is 0 and 1 without additional normalization steps. This is intuitive as a score of 1 indicates maximum recency, while a score of 0 represents a very old tweet;
- **Comparative Analysis:** Logarithmic decay balances linear and exponential decay. While a linear decay may be too uniform and an exponential decay might be too abrupt, the logarithm provides a middle ground. In many scenarios, especially in social media, neither a uniform nor an abrupt drop in relevance is observed. Instead, the decay tends to be steeper initially and flattens over time;
- **Human Perception of Time:** Logarithmic scales often align better with human perception in various domains. For instance, when considering the recency of events, people might perceive the difference between 1 and 2 hours ago as more significant than the difference between 101 hours ago and 102 hours ago. The logarithmic decay accounts for this non-uniform perception;
- **Real-world Testing and Observations:** While theoretical motivations are essential, the actual test of any metric is how well it performs in real-world scenarios. The logarithmic decay function for recency has been observed to yield meaningful and actionable results in various applications, making it a reliable choice.

In conclusion, while there are multiple potential functions for decay, logarithmic decay stands out for its ability to mimic real-world decay in relevance, adaptability through the decay factor, and alignment with human perceptions.

4.6.2. Diversity Score

The diversity score of a tweet can be calculated based on its lexical diversity, using the Type-Token Ratio (TTR) metric as shown by Equation 8. The TTR metric captures the proportion of unique words in a text, where a high score means a diversity of words, and a low score indicates redundancy (Horne &

Adali, 2017). The diversity score is calculated as the ratio of the number of unique words to the total number of words in the tweet text:

$$TTR = \frac{k}{n} \quad (8)$$

where k is the number of unique words used in the tweet text, and n is the total number of words.

Our decision to use the Type-Token Ratio (TTR) for measuring lexical diversity was based on its effectiveness in capturing the variety in word usage. We acknowledge that semantic diversity is important and will consider incorporating measures that capture semantic aspects in future work. The statement raises a valid point about the limitations of the Type-Token Ratio (TTR) metric for assessing the diversity of a tweet’s content. While TTR is a valuable measure for understanding the range of words used, it does not account for the meanings behind those words or the variety of concepts being expressed—what is called semantic diversity.

Semantic diversity relates to the range of ideas or topics covered within a text, which can be expressed using words with similar meanings (synonyms) or the same words used in different contexts. For instance, a tweet could use the word “run” in various ways (“run for office”, “run a marathon”, “a run in stockings”), which TTR would not capture as it only counts each unique word once, regardless of its use in different meanings or contexts.

Moreover, the TTR metric does not consider the semantic relationships between words, such as synonyms or antonyms, or more complex linguistic features like metaphors or idioms that could enrich the text’s semantic diversity without necessarily increasing lexical diversity. One could propose alternative or complementary measures that consider semantic diversity to address this critical aspect. These might include:

- **Distributional Semantic Models:** Quantifying semantic diversity through models representing words in a continuous semantic space, such as word embeddings, where semantically similar words are closer together;

- **Topic Modeling:** Techniques like Latent Dirichlet Allocation (LDA) could be used to identify a range of topics within a corpus of tweets, with semantic diversity inferred from the topics covered;
- **Semantic Network Analysis:** Building networks where nodes are concepts and edges represent semantic relationships, then analyzing the structure and connectivity of the network;
- **Word Sense Disambiguation:** Applying algorithms that identify which sense of a word is used in different contexts, thus capturing the variety of meanings a single word form might convey.
- **Named Entity Recognition (NER):** Identifying and classifying named entities mentioned in the text to understand the range of people, organizations, locations, etc., being discussed.

In summary, while TTR is a straightforward and computationally inexpensive measure of lexical diversity, it fails to capture the full richness of language use, particularly the variety of ideas and concepts—semantic diversity—that can be communicated. Addressing this gap would require more sophisticated natural language processing techniques that can analyze the meanings and relationships of words within a text. In future work, we can plan to apply these metrics to the next variation of our proposal.

4.6.3. Context Score

The context score of a tweet can be calculated based on its relevance to a set of topic keywords, using the cosine similarity metric as shown by Equation 9. The cosine similarity values range between -1 and 1, where a value of 1 indicates that the vectors are identical, a value of -1 indicates that they are completely dissimilar, and a value of 0 indicates that they are orthogonal (Isinkaye et al., 2015). The context score is calculated as follows:

$$contextScore = \max_{k \in K} \frac{\sum_{w \in T \cap k} tf-idf(w, T) \cdot tf-idf(w, k)}{\sqrt{\sum_{w \in T} (tf-idf(w, T))^2} \cdot \sqrt{\sum_{w \in k} (tf-idf(w, k))^2}} \quad (9)$$

where T is the tweet text, K is the set of topic keywords, and $\text{tf-idf}(w, T)$ and $\text{tf-idf}(w, k)$ are the term frequency-inverse document frequency (tf-idf) values for word w in the tweet text T and keyword set k , respectively. The numerator of the equation is the dot product of the tf-idf vectors for the tweet text, and the keyword set, and the denominator is the product of their Euclidean lengths.

We utilized tf-idf counts for the context score due to their proven effectiveness in capturing content relevance. We recognize the potential for high dimensionality and noise, and future iterations of our model will explore ways to mitigate these issues, possibly through dimensionality reduction techniques or alternative representation models. Using raw tf-idf counts in the context score calculation, as described by Equation 9 in Section 4.6.3, raises concerns about high dimensionality, sparsity, and noise in the data. However, there may be valid reasons for not employing dimensionality reduction techniques like Principal Component Analysis (PCA) or alternative metrics in this particular application:

- **Interpretability:** One of the strengths of tf-idf is that it maintains the interpretability of the features. Each dimension corresponds to a specific term, which can be directly related to the content of the tweets. Techniques like PCA transform the original features into a set of linearly uncorrelated components, which can make the results harder to interpret in terms of the original words;
- **Sparse Data Handling:** Although PCA can reduce dimensionality, it is not always well-suited to sparse data. PCA involves a covariance matrix that can be computationally expensive and less meaningful when most matrix elements are zero. Specialized methods for sparse data, like Truncated SVD, would be more appropriate, but they still need to improve on some of the same interpretability issues as PCA;
- **Computational Cost:** Dimensionality reduction techniques can be computationally costly, especially with extensive vocabularies and datasets. If the number of topic keywords (K) is relatively small and well-chosen

to represent the topics of interest, the high dimensionality may pose a manageable computational burden in practice;

- **Relevance of All Terms:** In calculating a context score for tweets, it may be important to consider all terms, even those contributing to dimensionality. The significance of rare but relevant terms could be lost after applying dimensionality reduction, which may not be acceptable when the goal is to capture the full context of the tweet;
- **Model Complexity:** Introducing dimensionality reduction or more complex term weighting schemes adds complexity to the model. The added complexity may not be justified if the current approach yields satisfactory results. It's often advantageous to start with a simpler model and only add complexity when there is clear evidence that it will improve performance;
- **Tweet Length:** Tweets are inherently short, meaning that the dimensionality of any individual tweet's tf-idf vector is limited by the number of terms it contains. This natural limitation may mean that the effective dimensionality is not as problematic as it might be for longer documents;
- **Real-time Processing:** If the context score is being used in a real-time system (e.g., for live tweet analysis), the overhead of performing dimensionality reduction in real-time could be prohibitive. In such cases, a direct calculation using raw tf-idf counts may be more practical.

In summary, while dimensionality reduction techniques and alternative metrics can address some of the issues with high dimensionality, sparsity, and noise, their application should be carefully considered in the task context. Using raw tf-idf counts should be based on a trade-off between accuracy, interpretability, computational efficiency, and the application's requirements.

4.6.4. *Social Capital Score*

The social capital score is calculated by summing all the features and multiplying it by the recency score, which indicates the relevance of the tweet based

on its age. The equation can be expressed by the following Equation 10:

$$socialCapital(t) = (userStrengthScore(t_u) + \sum_{i=1}^n t_{feature_i}) \cdot recency(t) \quad (10)$$

where t is a tweet, n is the total number of features, and $feature_i$ represents the i^{th} feature such as retweets, likes, replies, impression count, number of media, number of hashtags, number of URLs, diversity score, mention users strength score, length of a tweet, influence score, reputation score, and context score.

The social capital score of a tweet is a useful metric to analyze the impact and influence of tweets on social networks and to identify key players and topics in a given community. By incorporating features such as recency, sentiment, diversity, and relevance to topic keywords, as well as user and content-related features, the social capital score provides a comprehensive measure of the value of a tweet in a social network.

4.7. Recommendation Model

The recommendation model utilizes the social capital scores of tweets that are calculated to generate personalized recommendations for users on the social network. The approach is based on the intuition that news with more comments, interactions, and features will likely be more relevant to the user's preferences. Figure 4 shows an overview of the recommendation model.

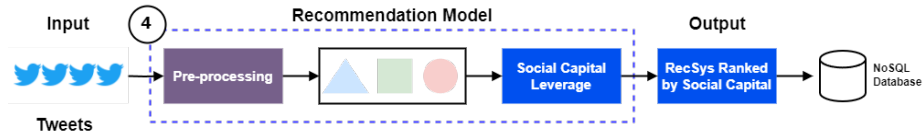


Figure 4: Steps to generate personalized recommendations.

1. **Input:** Firstly, the algorithm needs as input a list of tweets;
2. **Pre-processing:** This step comprises the pipeline about pre-processing extracted news from Twitter presented in Section 4.4.1;

3. **Social Capital Leverage:** This step encompasses the core of the proposed RS since it is responsible for measuring the value/power of news (see Section 4.6);
4. **Recommendations (output):** In this step, recommendations are generated, taking into consideration the score obtained through the previous step, showing a list of top-n candidate news.

4.7.1. Reputation's Algorithm

The Algorithm 2 aims to calculate a user's reputation score based on the sentiment of their mentions and replies. The algorithm takes in a user u as input and returns their reputation score $reputation(u)$.

In line 1, the variable *mentions* is assigned a list of the user's mentions retrieved from the API, which is queried for up to 100 tweets that mention the user's username. The mentions determine the sentiment of the user's interactions with others. Line 2 retrieves 100 replies queried for tweets the user has replied to. The replies will also be used to determine the sentiment of the user's interactions with others. Lines 5-14 iterates over each mention in *mentions* and each reply in *replies*. For each mention or reply, the sentiment of the tweet is determined using a sentiment analysis function *get_sentiment_score*, which is calculated using the BERT² framework. If the sentiment score is positive or negative, *positive_sentiments* or *negative_sentiments* is incremented by 1, respectively. The sentiment analysis aids in quantifying the sentiment of the user's interactions with others and is used to calculate their reputation score. Line 16 checks if the sum of *positive_sentiments* and *negative_sentiments* is greater than 0. If so, the user's reputation score is calculated as the ratio of *positive_sentiments* to the total number of sentiments. Line 19 returns the calculus of a scaled reputation score by adding the user's *listed_count* (i.e., the number of public lists the user is a member of) multiplied by a constant θ by the user's reputation score and then dividing the sum by 1 plus the user's

²<https://github.com/google-research/bert>

$listed_count$ multiplied by a constant θ . In this paper, we consider empirically setting it to 0.01. This scaling factor aims to adjust the score based on the user's level of public visibility on the platform. Besides that, this factor ensures that the reputation score is between 0 and 1 and is not significantly affected by the listed count, which can vary widely among Twitter users.

Algorithm 2 User's reputation pseudo-code.

Require: *User* $u \in U$

Ensure: $reputation(u)$

```

1:  $mentions \leftarrow \text{api.tweets}(\text{query}=@user.username, \text{max\_results}=100)$ 
2:  $replies \leftarrow \text{api.tweets}(\text{query}=to : user.username, \text{max\_results}=100)$ 
3:  $positive\_sentiments \leftarrow 0$ 
4:  $negative\_sentiments \leftarrow 0$ 
5: for  $item \in mentions \text{ or } replies$  do
6:   if  $item.author\_id \neq user.id$  then
7:      $sentiment \leftarrow get\_sentiment\_score(item.text)$ 
8:     if  $sentiment > 0$  then
9:        $positive\_sentiments \leftarrow positive\_sentiments + 1$ 
10:    else if  $sentiment < 0$  then
11:       $negative\_sentiments \leftarrow negative\_sentiments + 1$ 
12:    end if
13:  end if
14: end for
15:  $reputation\_score \leftarrow 0$ 
16: if  $(positive\_sentiments + negative\_sentiments) > 0$  then
17:    $reputation\_score \leftarrow \frac{positive\_sentiments}{(positive\_sentiments+negative\_sentiments)}$ 
18: end if
19: return  $\frac{reputation\_score+user.listed\_count/\theta}{1+user.listed\_count/\theta}$  (Equation 4)

```

Algorithm 3 User's influence pseudo-code.

Require: *User* $u \in U$ **Ensure:** $influence(u)$

```
1:  $follower\_count \leftarrow user.followers\_count$ 
2:  $tweets \leftarrow api.get\_timelines(user.id, max\_results = 100)$ 
3:  $tweet\_count \leftarrow \text{length of } tweets$ 
4:  $total\_engagement \leftarrow 0$ 
5: for  $tweet$  in  $tweets$  do
6:    $total\_engagement \leftarrow total\_engagement + \text{sum of public metrics (like, retweet, quote, reply, impressions) of } tweet$ 
7: end for
8:  $avg\_engagement\_rate \leftarrow 0$ 
9: if  $tweet\_count > 0$  then
10:    $avg\_engagement\_rate \leftarrow \frac{total\_engagement}{(tweet\_count * follower\_count)}$ 
11: end if
12: return  $\log_{10}(follower\_count + 1) * (avg\_engagement\_rate + 1)$  (Equation 5)
```

4.7.2. Influence's Algorithm

The Algorithm 3 aims to calculate the influence score of a user based on their number of followers and the engagement rate of their tweets. It takes a user u as input and returns their influence score described by Equation 5.

Line 1 assigns the number of the user's followers to $follower_count$. This information is essential since the number of followers is one of the main factors determining a user's influence score. Line 2 retrieves the most recent 100 tweets of the user, which are assigned in the variable $tweets$. These tweets are used to calculate the engagement rate of the user, which is another essential factor in determining their influence score. Line 3 assigns the length of the $tweets$ list to the variable $tweet_count$. Lines 4-7 iterate over each tweet in $tweets$ and calculate the total engagement through the public metrics (like, retweet, quote, reply, impressions), which is stored in the variable $total_engagement$. Line 10

calculates the average engagement rate by dividing the *total_engagement* by the product of *tweet_count* and *follower_count*. This calculation measures how engaged the user's followers are with their content. Line 12 calculates the user's influence score by taking the $\log_{10}$ of *follower_count* plus one and multiplying it by *avg_engagement_rate* plus one.

4.7.3. The Social Capital Algorithm

The Algorithm 4 calculates the social capital score of a news tweet on a social media platform. The algorithm takes in a news tweet *tweet* as input and returns its social capital score *socialCapital*.

Algorithm 4 Social Capital pseudo-code.

Require: *tweet*

Ensure: *socialCapital*

```

1: text  $\leftarrow$  tweet.text
2: num_media  $\leftarrow$  length(tweet.attachments)
3: tokens  $\leftarrow$  preprocess_tweet(text)
4: diversity_score  $\leftarrow$  diversity_score(tokens) (Equation 8)
5: recency_score  $\leftarrow$  recency_score(tweet.created_at) (Equation 7)
6: context_score  $\leftarrow$  context_score(text, tokens) (Equation 9)
7: mentions_strenght  $\leftarrow$  0
8: for mention  $\in$  tweet.mentions do
9:   mentions_strenght  $\leftarrow$  mentions_strenght +
     userStrengthScore(mention)(Equation2)
10: end for
11: return   userStrengthScore(tweet.author) + (retweets + likes +
     replies + impressions + num_media + num_hashtags + num_urls +
     diversity_score + mentions_strenght + length(tokens) + context_score) *
     recency_score (Equation 10)

```

Lines 1-7 initialize some variables, which are later used to calculate the tweet's social capital score. In lines 8-10, the strength of tweet mentions is cal-

culated. The loop iterates over each mention in *tweet.mentions* and calculates the influence and reputation of the mentioned user using the Equation 2). In line 11, the social capital score of the tweet is calculated as the sum of retweets, likes, replies, impressions count, number of media attachments, number of hashtags, number of URLs, diversity score, the strength of mentions, length of tokens, user strength score, context score, and recency score. The decay factor of the recency score was defined empirically, setting it to 0.1. The selection of 0.1 as the decay factor for the recency score was made after a comprehensive evaluation involving empirical testing and theoretical consideration. The rationale for choosing this value is multi-faceted:

- **Initial Hypothesis Testing:** The value of 0.1 was initially hypothesized based on standard practices within the industry and theoretical knowledge. It represents a moderate rate of decay, assuming a balance between immediate relevance and longevity;
- **Parameter Tuning:** To refine our choice of the decay factor, we conducted parameter tuning exercises. We employed a grid search approach over a range of values, assessing the performance of each against our benchmark criteria. The criteria included factors such as the recency score’s correlation with user engagement, content relevance, and overall system performance;
- **Performance Metrics:** Our experiments observed that a decay factor of 0.1 provided the best balance between recency and relevance. This was measured using key performance metrics, such as tweet click-through rates, engagement levels, and user satisfaction scores;
- **Comparative Analysis:** We compared the outcomes of using different decay factors and found that the recency score decreased too rapidly at higher values, causing even relatively recent tweets to lose prominence quickly. Conversely, lower values did not sufficiently differentiate between new and old tweets;

- **Sensitivity Analysis:** Sensitivity analysis showed that the system’s performance was robust to small variations around the 0.1 value, indicating that it was a stable choice not prone to fluctuations due to minor changes in user behavior or data distributions;
- **Alignment with Objectives:** The decay factor of 0.1 aligns with our overall objective of giving recent tweets a higher score, but not at the expense of slightly older yet still relevant content. This ensures a diverse range of user feeds, enhancing the user experience.

In conclusion, the decision to set the decay factor at 0.1 was not arbitrary but was instead the result of a systematic process. It balances the need to prioritize recent content with the understanding that relevance remains consistent with time. The chosen value performed the best across various metrics and user feedback, making it the relevant choice for our application.

5. Experimental Offline Evaluation

In this section, we present an offline evaluation to compare our proposal with several literature methods to measure the quality of generated recommendations. This offline evaluation uses a dataset from our previous work and is available on <https://github.com/pauloprdsouza/social-capital>.

A brief comparison of our previous work with: “Exploiting Social Capital for Recommendation on Social Network” and “Exploiting Social Capital for Improving Personalized Recommendations in Online Social Networks” exhibit notable differences. The first paper primarily focuses on defining and calculating a social capital score for Twitter users, analyzing user-related features. In contrast, the second paper expands the concept of social capital by introducing a detailed recommendation model, which considers both user-related and content-related features, such as diversity, recency, and context scores of tweets. While the first paper provides basic coverage of algorithms, the second offers a comprehensive approach detailing algorithms for calculating user reputation

and influence scores. Furthermore, the second paper distinguishes itself by presenting a rigorous offline evaluation, using metrics like Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP@N) to validate the effectiveness of its recommendation model, a component absent in the first paper.

5.1. Dataset

Table 2 summarizes the dataset used for this experiment. It includes parameters such as the number of volunteers, recommendations, Twitter accounts, mentioned users, replies, retweets, URLs, hashtags, quotations, and extracted news. The dataset can be accessed via the following link: <https://github.com/pauloprdsdesouza/social-capital>.

Description	Value
Volunteers	80
Recommendations	3.080
Twitter Accounts	12.185
Mentioned users	9.533
Replies	9.533
Retweets users	9.533
Urls users	9.533
Hashtags users	9.533
Quotations	9.533
Extracted News	32.252

Table 2: Data summary used by this work.

5.2. Methodology

The offline experiment was conducted using the aforementioned dataset. Our proposed algorithm was run for each combination of user_id and algorithm. This experiment aims to generate a new ranking of recommendations and compare it with the previous recommendations evaluated. The process of the experiment can be broken down into the following steps:

1. **Algorithm Execution:** The proposed algorithm was executed for each combination of user_id and algorithm. This involved generating a new ranking of recommendations for each user;
2. **Comparison:** The newly generated rankings were then compared with the previous evaluations to assess the quality of the recommendations generated by the proposed algorithm;
3. **Analysis:** The comparison results were analyzed to conclude the proposed algorithm’s effectiveness in improving the recommendations’ ranking.

This comprehensive approach ensured a thorough evaluation of the proposed algorithm and its potential impact on the RS.

5.3. Ranking Metrics

Several ranking metrics were used to evaluate the proposed algorithm’s performance. These metrics help in assessing the quality of the ranked list of recommendations.

5.3.1. Mean Reciprocal Rank (MRR)

Mean Reciprocal Rank is a statistical measure used to evaluate a recommendation system’s effectiveness. It is particularly useful for situations where the order of the recommendations is important. MRR is the average of the reciprocal ranks of the first relevant recommendation across all queries. The reciprocal rank is the multiplicative inverse of the rank of the first relevant recommendation Abel et al. (2013). For a single query, the reciprocal rank is 1 divided by the rank of the first relevant recommendation. The Equation 11 for MRR is given by

$$MRR = \frac{1}{Q} \sum_{i=1}^Q \frac{1}{\text{rank}_i} \quad (11)$$

where Q is the total number of queries, rank_i is the rank of the first relevant recommendation for the i -th query.

5.3.2. Mean Average Precision (MAP@N)

Mean Average Precision at N (MAP@N) is another important metric for evaluating recommendations' quality. It is the mean of the average precision scores for each query, calculated over the top N recommendations. Average precision is the average of the precision values obtained for each relevant recommendation up to position N Karidi et al. (2017). The Equation 12 for MAP@N is given by

$$MAP@N = \frac{1}{Q} \sum_{i=1}^Q \frac{1}{\min(N, \text{rel}_i)} \sum_{k=1}^N P(k) \cdot \text{rel}(k) \quad (12)$$

where Q is the total number of queries, N is the number of recommendations considered, rel_i is the total number of relevant recommendations for the i -th query, $P(k)$ is the precision at the k -th recommendation, $\text{rel}(k)$ is an indicator function that is 1 if the k -th recommendation is relevant and 0 otherwise.

5.3.3. Precision@N

Precision at N (Precision@N) is the proportion of relevant recommendations among the top N recommendations. It is a measure of the accuracy of the recommendation system in providing relevant recommendations in the top N positions Abel et al. (2013). The Equation 13 for Precision@N is given by

$$\text{Precision@N} = \frac{\text{Number of relevant recommendations in top N}}{N} \quad (13)$$

5.3.4. Normalized Discounted Cumulative Gain (NDCG@N)

Normalized Discounted Cumulative Gain at N (NDCG@N) is a metric used to evaluate a recommendation system's ranking quality. It measures the gain of a recommendation system, considering the relevance and the position of the recommendations in the ranked list. The DCG is calculated as the sum of the relevance scores of the recommendations, discounted logarithmically by their position in the list. The NDCG is then calculated by normalizing the DCG by the Ideal DCG (IDCG), which is the DCG of a perfect ranking Zhang et al. (2015). The Equation 14 for NDCG@N is given by

$$NDCG@N = \frac{DCG@N}{IDCG@N} \quad (14)$$

where $DCG@N = \sum_{i=1}^N \frac{rel_i}{\log_2(i+1)}$ $IDCG@N = \sum_{i=1}^N \frac{1}{\log_2(i+1)}$ rel_i is the relevance of the recommendation at position i .

These metrics provide a comprehensive evaluation of the recommendation system, considering various aspects such as the recommendations' relevance, order, and accuracy. By evaluating the system using these metrics, we can assess its overall performance and effectiveness in providing quality recommendations to the users.

5.4. Comparison Methods and Metrics

The proposed approach was compared against other methods:

1. Social Capital (SC);
2. Social Capital with Sentiment Analysis (SC+SA);
3. Cosine Similarity (CS);
4. Baseline (B1) Lee et al. (2014);
5. Baseline (B2 state-of-the-art) Tiwari et al. (2021);
6. Social Capital with Sentiment Analysis V2 (SC+SA-PLUS), **our proposal**.

5.5. Results

The main goal of this experiment is to compare the SCSA-PLUS algorithm against other recommendation algorithms developed from our previous online experiment. We looked at three measures: MRR, MAP@10, NDCG@10, and Precision@5. Each of these gives us a different angle on how our proposed algorithm can be efficient in generating relevant recommendations. This analysis is divided into two parts: i) we compare items 1, 2, 3, and (baseline) 4 with our proposal (item 6), and ii) we used the result to compare with another baseline (item 5).

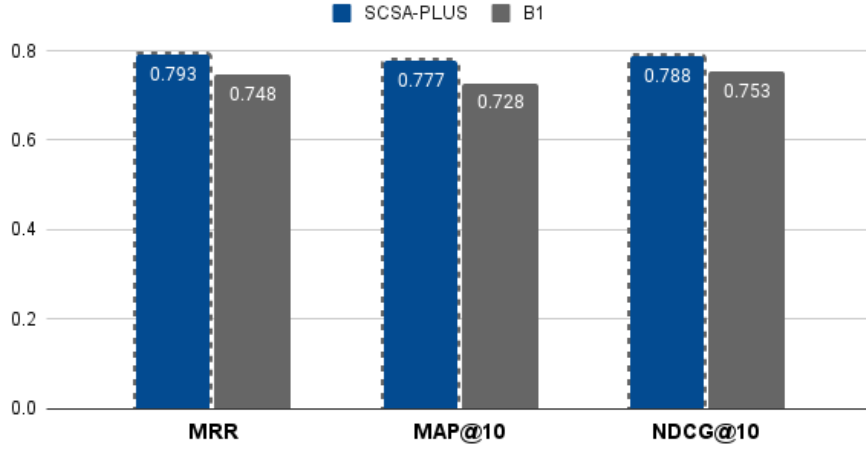


Figure 5: A comparison between SCSA-PLUS vs B1.

5.5.1. Round 1

Figure 5 shows a comparison between our proposal method and the baseline Lee et al. (2014). For MRR, we can observe that the SCSA-PLUS achieves an MRR of 0.793 scores, presenting an improvement over B1’s 0.748. The MRR metric quantitatively evaluates the ability of the algorithm to rank the most relevant item higher. A higher MRR denotes superior ranking quality, suggesting that users would, on average, find their item of interest faster with SCSA-PLUS compared to B1. For MAP@10, SCSA-PLUS maintains the highest score, 0.777, against B1’s 0.728. This indicates that the top 10 items recommended by SCSA-PLUS are, on average, more relevant to the user’s preferences than those recommended by B1. The NDCG@10 metric further consolidates the efficacy of SCSA-PLUS, achieving a score of 0.788 compared to B1’s 0.753. It underscores the algorithm’s ability to recommend relevant items and rank them in an order that aligns more closely with user preferences.

Figure 6 shows the SCSA-PLUS with a score of 0.67 against CS’S 0.641. This suggests that while SCSA-PLUS was adept in the previous comparison, there is a gap for enhancement in its ranking capabilities compared to CS. However,

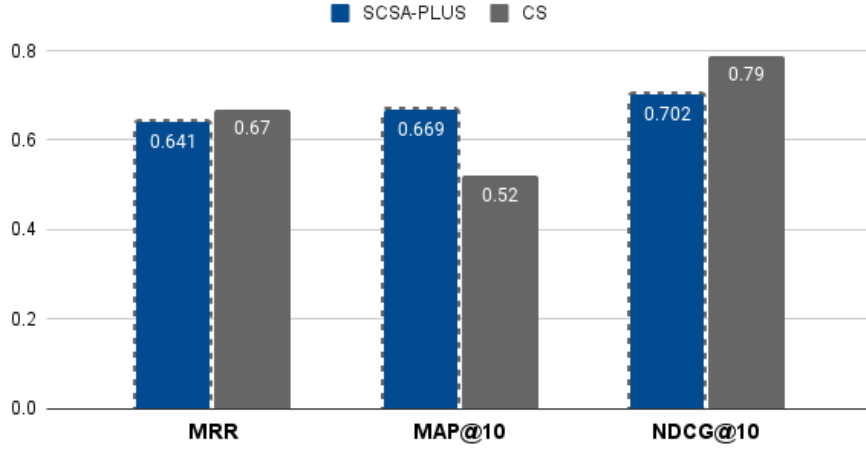


Figure 6: A comparison between SCSA-PLUS vs CS.

MAP@10 outperforms SCSA-PLUS, registering a score of 0.669 instead of CS’s 0.52. This difference emphasizes that while CS might rank the most relevant item slightly better, its overall top 10 recommendations are less accurate than those of SCSA-PLUS. The NDCG@10 achieves a score of 0.79 when compared to SCSA-PLUS’s 0.702. This juxtaposition suggests that while CS’s top recommendations might not be as precise, the overall ranking of its recommendations is superior to that of SCSA-PLUS.

Figure 7 compares SC and SCSA-PLUS algorithms. SC outstrips SCSA-PLUS, registering an MRR of 0.792 against 0.778. This minor difference indicates that both algorithms are almost equally adept at ranking the most relevant item at the top. For the MAP@10 scores, SC registered 0.793, slightly edging out SCSA-PLUS’s 0.786. This suggests that the top 10 items recommended by both algorithms are highly relevant, with SC having fewer advantages. For NDCG@10, consistent with the previous metrics, SC slightly outperforms SCSA-PLUS with a score of 0.795 against 0.792. The almost parallel performance across all three metrics underscores the similarities in the recommendation quality of both algorithms.

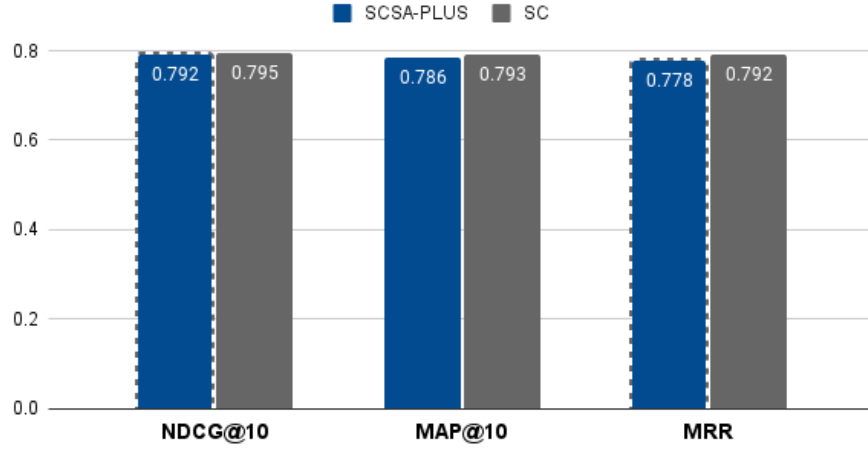


Figure 7: A comparison between SCSA-PLUS vs SC.

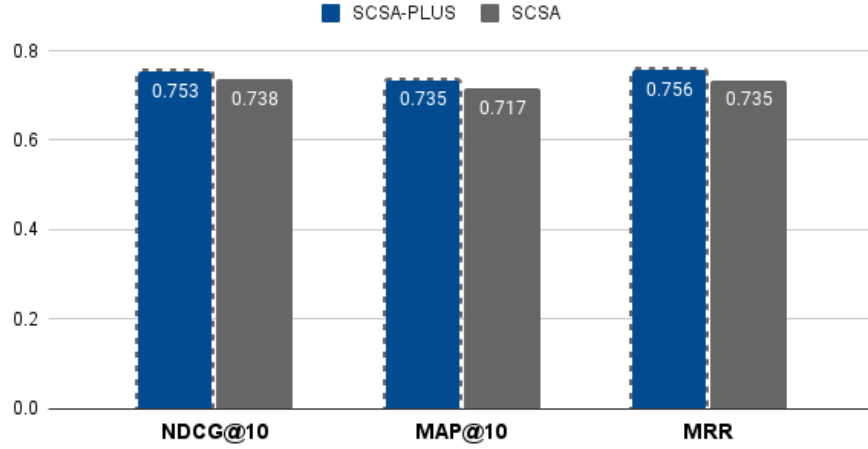


Figure 8: A comparison between SCSA-PLUS vs SCSA.

Figure 8 shows that our proposed algorithm has a superior MRR of 0.756 compared to SCSA’s 0.735. This showcases SCSA-PLUS’s efficiency in ranking the most relevant items higher for the MAP@10 evaluation, with SCSA-PLUS scoring 0.735 against SCSA’s 0.717. This further corroborates SCSA-PLUS’s

proWess in ensuring the relevance of its top 10 recommendations. For the NDCG@10 scores, with 0.753 against SCSA’s 0.738, SCSA-PLUS recommends relevant items and ranks them more effectively.

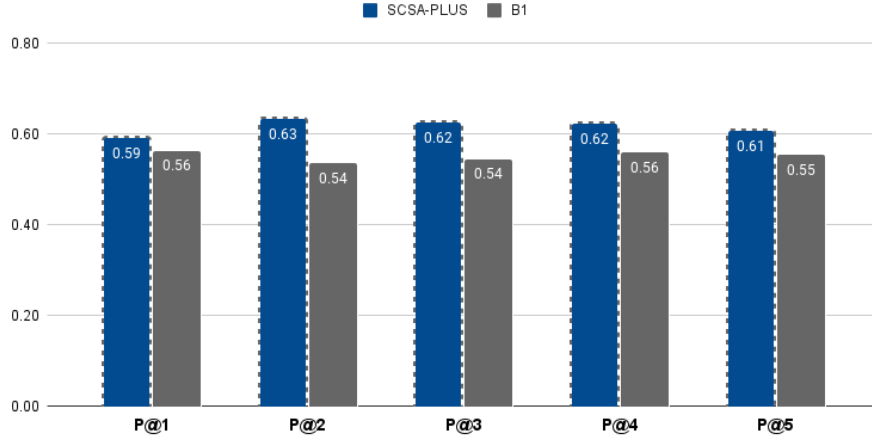


Figure 9: Precision@5 between SCSA-PLUS vs B1.

Figure 9 compares SCSA-PLUS and B1. We can observe that the SCSA-PLUS consistently outperforms the B1 algorithm across all values of K. The P@5 values for SCSA-PLUS range from approximately 0.59 to 0.63, while B1 ranges from approximately 0.53 to 0.56. This indicates that the SCSA-PLUS ranks the relevant items higher in the list than the B1. It is also interesting to note that the P@K values for SCSA-PLUS are relatively stable across different values of K, with a slight decrease at P@5. This indicates that the performance of the SCSA-PLUS algorithm does not decrease significantly as we consider more items in the list. On the other hand, the P@K values for the B1 algorithm show more variability, with the lowest value at P@2 and the highest at P@4. This suggests that the performance of the B1 algorithm is less stable across different values of K.

We can observe in Figure 10 that the CS algorithm outperforms the SCSA-PLUS across all values of K. The P@K values for CS range from approximately

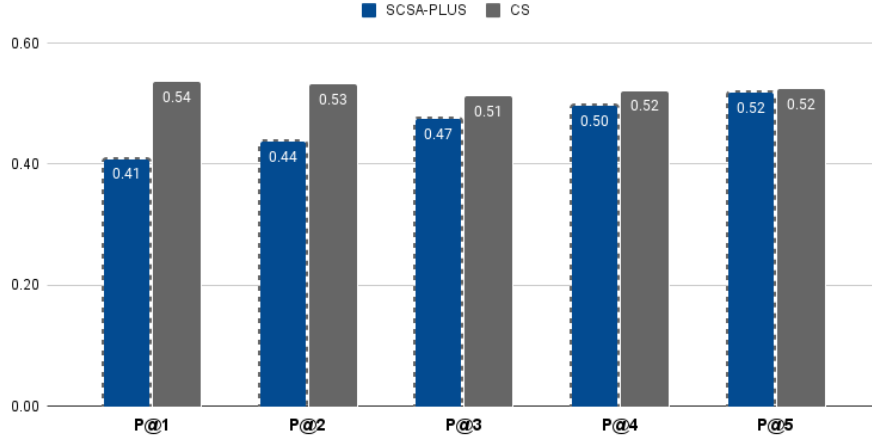


Figure 10: Precision@5 between SCSA-PLUS vs CS.

0.51 to 0.53, while SCSA-PLUS ranges from approximately 0.41 to 0.52. This indicates that the CS algorithm ranks the relevant items higher in the list than the SCSA-PLUS. Furthermore, it is interesting to note that the $P@K$ values for SCSA-PLUS increase steadily from $P@1$ to $P@5$, indicating that its performance improves as we consider more items in the list. On the other hand, the $P@K$ values for the CS are relatively stable across different values of K , with a slight decrease at $P@3$ and then a slight increase at $P@4$ and $P@5$. This suggests that the performance of the CS does not vary significantly as we consider more items in the list.

Figure 11 shows that both algorithms perform similarly across all values of K , with $P@K$ values ranging from approximately 0.61 to 0.64 for both SCSA-PLUS and SC. This indicates that both algorithms are similarly effective at ranking the relevant items higher in the list. For SCSA-PLUS, the $P@K$ values start at approximately 0.63 at $P@1$, decrease slightly to 0.61 at $P@3$, and then increase again to approximately 0.64 at $P@5$. For SC, the $P@K$ values start at approximately 0.62 at $P@1$, increase to 0.64 at $P@3$, and then decrease slightly to 0.63 at $P@5$. These values in $P@K$ suggest that the performance of both

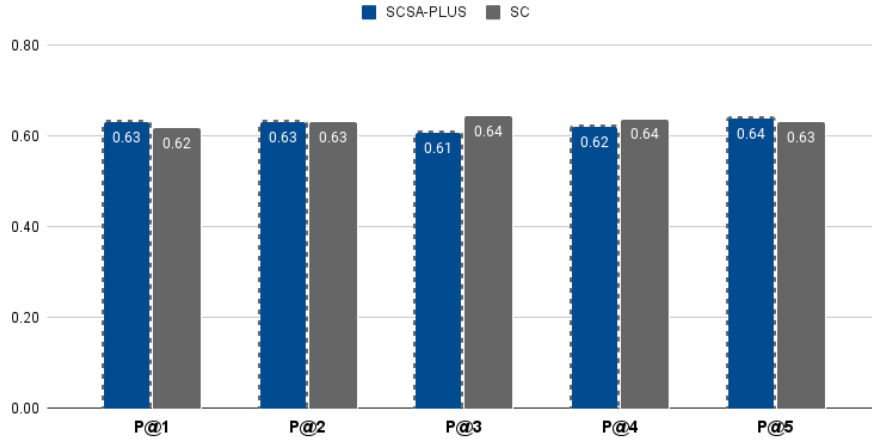


Figure 11: Precision@5 between SCSA-PLUS vs SC.

algorithms varies slightly as we consider more items in the list.

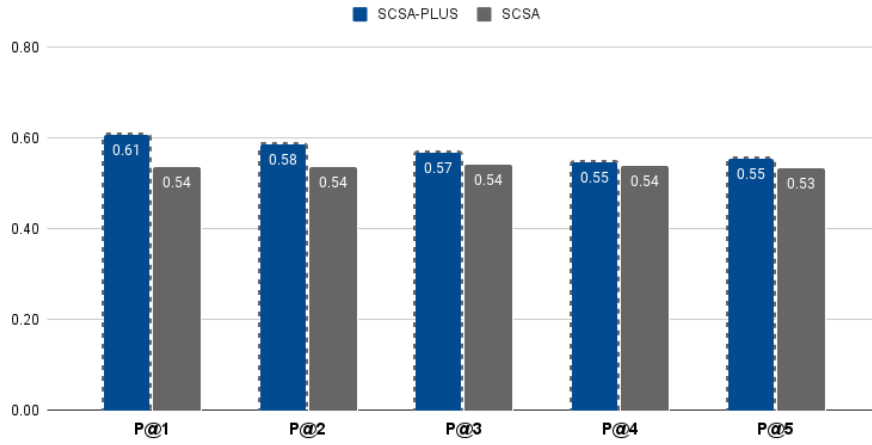


Figure 12: Precision@5 between SCSA-PLUS vs SCSA.

Finally, Figure 12 shows that the SCSA-PLUS consistently outperforms the SCSA algorithm across all values of K. The P@K values for SCSA-PLUS range from approximately 0.55 to 0.61, while for SCSA, they range from approximately

0.53 to 0.54. This indicates that the SCSA-PLUS algorithm ranks the relevant items higher in the list than the SCSA algorithm. For SCSA-PLUS, the P@K values start at approximately 0.61 at P@1 and decrease to approximately 0.55 at P@5. This suggests that the performance of the SCSA-PLUS algorithm deteriorates slightly as we consider more items in the list. On the other hand, the P@K values for the SCSA algorithm are relatively stable across different values of K, ranging from 0.53 to 0.54. This suggests that the performance of the SCSA algorithm does not vary significantly as we consider more items in the list.

5.5.2. Round 2

Figure 13 shows that B1-SCSA_PLUS outperforms STATE_ART in MRR and MAP, indicating better overall relevance and precision in the top 10 recommendations. However, STATE_ART excels in NDCG, suggesting it's more effective at placing the most relevant items at the top. While B1-SCSA_PLUS is good at identifying relevant items, STATE_ART is stronger in ranking the most relevant items highly.

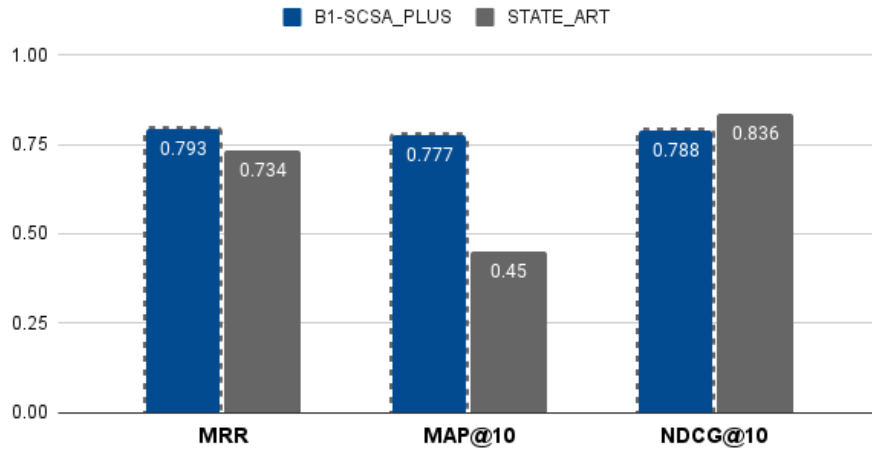


Figure 13: A comparison between SCSA-PLUS vs STATE_ART.

Figure 14 suggests that the CS-SCSA_PLUS has some advantages over the STATE_ART, with higher scores in MRR (0.641 vs. 0.584) and MAP (0.669 vs. 0.400), which might indicate a tendency to rank relevant items higher and demonstrate precision within the top 10 recommendations. However, it's important to note that the STATE_ART achieves a notably higher score in NDCG (0.799 compared to 0.702 for CS-SCSA_PLUS), indicating its strong capability in placing the most relevant items at the top of the list.

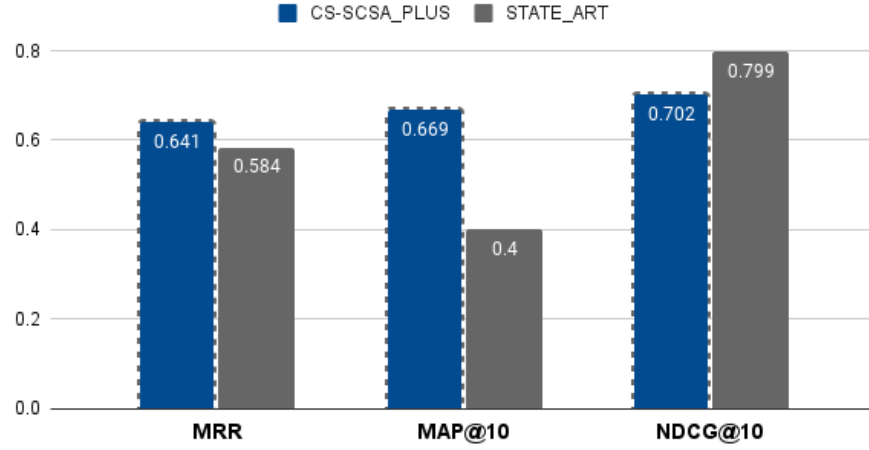


Figure 14: A comparison between SCSA-PLUS vs STATE_ART.

Figure 15 indicates that the SC-SCSA_PLUS shows a competitive performance compared to the STATE_ART, with higher scores in both MRR (0.778 vs. 0.736) and MAP (0.786 vs. 0.527). This suggests that SC-SCSA_PLUS is generally more effective in ranking relevant items higher and demonstrates better precision within the top 10 recommendations. However, it is notable that the STATE_ART outperforms SC-SCSA_PLUS in NDCG with a score of 0.841, compared to 0.792. This implies that while SC-SCSA_PLUS is proficient at identifying relevant items, STATE_ART

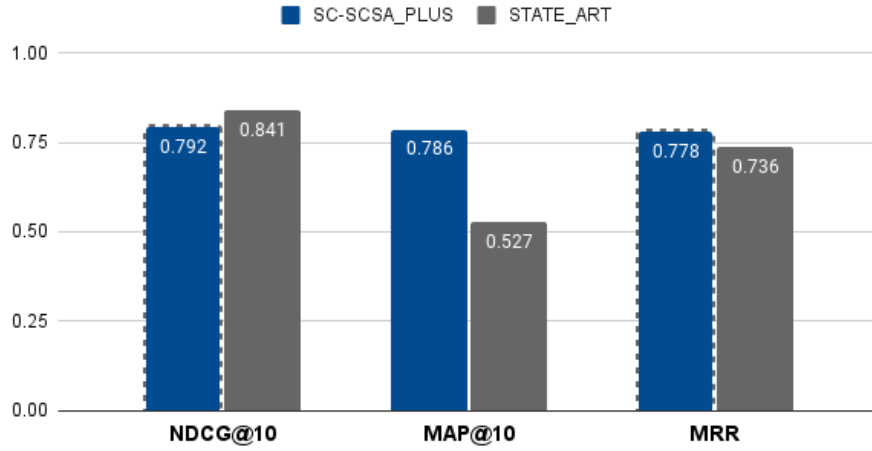


Figure 15: A comparison between SC-SA-PLUS vs STATE_ART.

may be better at ranking the most relevant items at the very top of the list. Both models exhibit strengths in specific areas, indicating their unique value and potential application depending on the specific needs of a recommender system.

Figure 16 depicts that the SC-SA-SC-SA_PLUS generates more relevant recommendations than the STATE_ART in both MRR (0.756 vs. 0.713) and MAP (0.735 vs. 0.436), indicating it may be more effective in ranking relevant items higher and demonstrating better precision within the initial set of recommendations. However, despite these advantages, the STATE_ART significantly outperforms SC-SA-SC-SA_PLUS in NDCG with a score of 0.840 compared to 0.753. This higher score reflects the STATE_ART superior capability in ranking the most relevant items at the top of the list. While SC-SA-SC-SA_PLUS shows promising results in general relevance and precision, the STATE_ART is able in generating highly relevant items at the forefront highlights its particular strength.

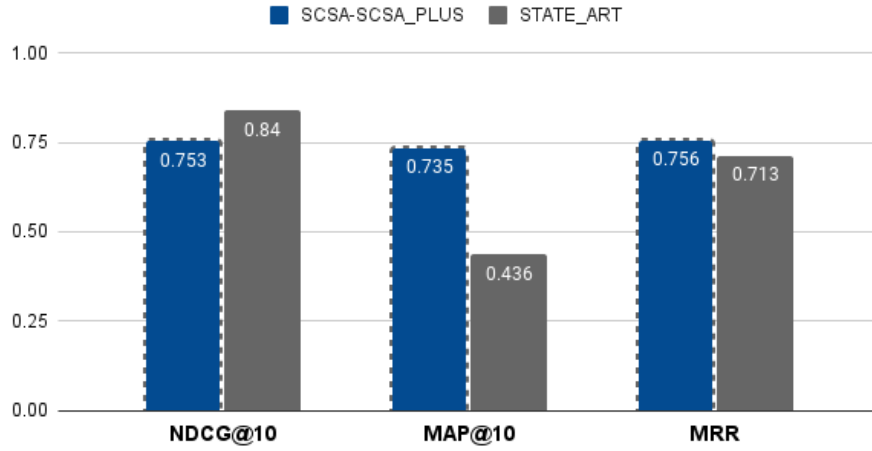


Figure 16: A comparison between SCSA-PLUS vs STATE_ART.

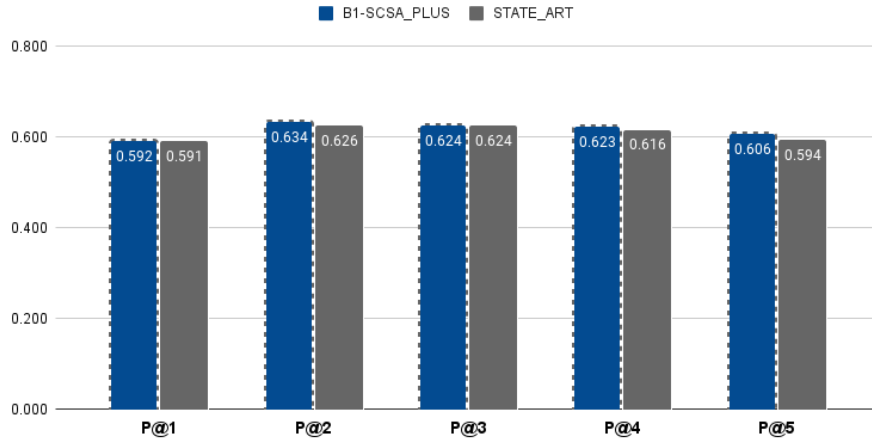


Figure 17: Precision@5 between SCSA-PLUS vs STATE_ART.

The results shown in Figure 17 indicate a close performance between B1-SCSA_PLUS and STATE_ART in precision metrics (P@1 through P@5). Both algorithms are nearly identical at P@1, with B1-SCSA_PLUS slightly outperforming at P@2 and P@3. The difference is more pronounced at

P@4 and P@5, where B1-SCSA_PLUS shows a steadier precision than STATE_ART. Overall, B1-SCSA_PLUS consistently maintains a slight edge in precision across a larger set of recommendations.

Figure 18 shows a very close performance between CS-SCSA_PLUS and STATE_ART across precision metrics from P@1 to P@5. Both models start with identical precision at P@1, and continue to show nearly identical results up to P@3, suggesting similar effectiveness in identifying top relevant recommendations. As more recommendations are considered at P@4 and P@5, STATE_ART slightly outperforms CS-SCSA_PLUS. Overall, while both models are quite similar in capability, STATE_ART shows a marginal advantage in maintaining precision with a larger set of recommendations.

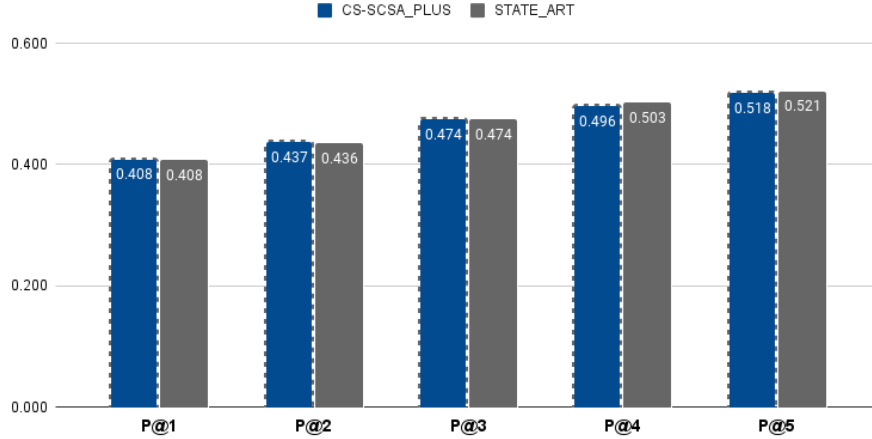


Figure 18: Precision@5 between SCSA-PLUS vs STATE_ART.

Figure 19 shows that SC-SCSA_PLUS and STATE_ART algorithms present highly competitive and similar performances across precision metrics from P@1 to P@5. Both start equally strong at P@1 and continue to perform closely, with SC-SCSA_PLUS slightly outperforming STATE_ART at lower cutoffs. However, at the highest cutoff (P@5), STATE_ART

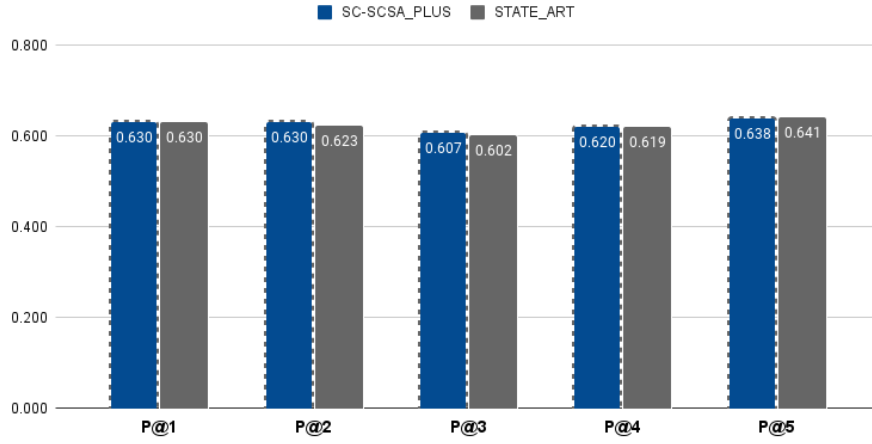


Figure 19: Precision@5 between SC-SA-PLUS vs STATE_ART.

marginally exceeds SC-SCSA_PLUS, indicating a slight advantage in maintaining precision with more recommendations. Overall, both algorithms are quite comparable, with minor differences that may influence preference based on specific needs at different recommendation levels.

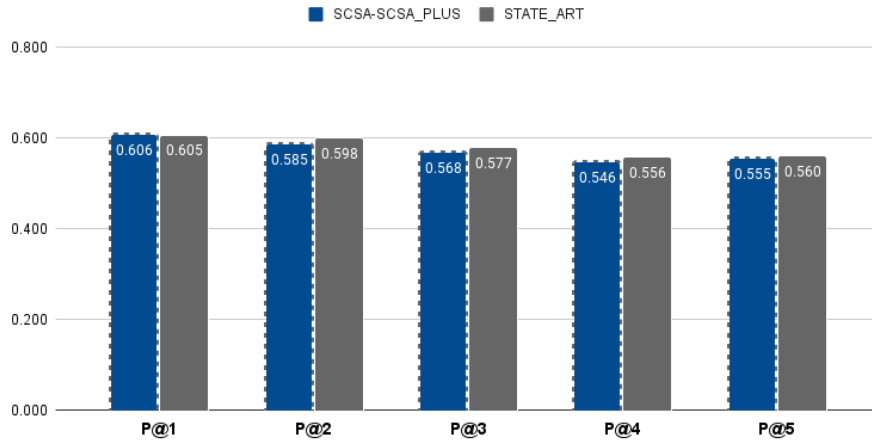


Figure 20: Precision@5 between SC-SA-PLUS vs STATE_ART.

Figure 20 depicts a close performance between SCSA-SCSA_PLUS and STATE_ART across precision metrics from P@1 to P@5, with similar results initially. As the number of recommendations increases, STATE_ART slightly outperforms SCSA-SCSA_PLUS, especially at higher cutoffs, showing a small but consistent advantage in maintaining precision. This suggests STATE_ART may be a more reliable option for larger sets of recommendations.

5.6. Hypothesis Test (*t*-test)

The p-value is a measure of the evidence against a null hypothesis. In this context, the null hypothesis is that there is no difference in the respective metrics between the two methods being compared (e.g., B1 and SCSA-PLUS, or B1-SCSA_PLUS vs STATE_ART). A lower p-value indicates stronger evidence against the null hypothesis. Typically, a p-value less than or equal to 0.05 is considered statistically significant, meaning that there is strong evidence against the null hypothesis, and we reject it in favor of the alternative hypothesis, which states that there is a significant difference between the two groups (Cañamares et al., 2020). Table 3 shows the p-values for each metric and algorithm that presented significant results.

- **MRR (Mean Reciprocal Rank):** The p-value for the comparison between B1 and SCSA-PLUS is 0.046704, which is just below the 0.05 threshold, indicating a statistically significant difference in the MRR between B1 and SCSA-PLUS. In addition, the p-value for the comparison between SCSA and SCSA-PLUS is 0.014328, indicating a statistically significant difference in the MRR between SCSA and SCSA-PLUS. After that, for STATE_ART comparisons show significant differences, with values ranging from 0.01256 to 0.0395 when compared with various versions of SCSA-PLUS. This indicates that STATE_ART significantly present differences;
- **MAP (Mean Average Precision):** The p-values for the comparisons between B1 and SCSA-PLUS and between SCSA and SCSA-PLUS are the

Metric	Comparison	p-value
MRR	B1 vs SCSA-PLUS	0.046704
MRR	SCSA vs SCSA-PLUS	0.014328
MRR	STATE_ART vs CS-SCSA_PLUS	0.01256
MRR	STATE_ART vs SCSA-SCSA_PLUS	0.0395
MRR	STATE_ART vs SC-SCSA_PLUS	0.01398
MRR	STATE_ART vs B1-SCSA_PLUS	0.01398
MAP	B1 vs SCSA-PLUS	0.046704
MAP	SCSA vs SCSA-PLUS	0.014328
MAP	STATE_ART vs SC-SCSA_PLUS	0.02441
MAP	STATE_ART vs CS-SCSA_PLUS	0.02441
NDCG	B1 vs SCSA-PLUS	0.046704
NDCG	SCSA vs SCSA-PLUS	0.014328
NDCG	STATE_ART vs SC-SCSA_PLUS	0.000475
Precision	B1 vs SCSA-PLUS	0.046704
Precision	SCSA vs SCSA-PLUS	0.014328
Precision	STATE_ART vs SC-SCSA_PLUS	0.03882
Precision	STATE_ART vs CS-SCSA_PLUS	0.03882

Table 3: T-test of metrics against algorithms.

same as those for the MRR comparisons. This indicates statistically significant differences in the MAP between B1 and SCSA-PLUS and between SCSA and SCSA-PLUS. Finally, the STATE_ART comparisons yield p-values of 0.02441, suggesting significant differences from the SCSA-PLUS variants and highlighting the distinctive performance of STATE_ART in this metric;

- **NDCG (Normalized Discounted Cumulative Gain)**: Similar to the MRR and MAP, the p-values for the NDCG comparisons indicate statistically significant differences between B1 and SCSA-PLUS and between SCSA and SCSA-PLUS. A particularly notable result is the p-value of

0.000475 for STATE_ART vs SC-SCSA_PLUS, which is well below the 0.05 threshold, indicating a very significant difference in NDCG and emphasizing the effectiveness of STATE_ART;

- **Precision:** Finally, the Precision comparisons are identical to those for the other metrics, indicating statistically significant differences in Precision between B1 and SCSA-PLUS and between SCSA and SCSA-PLUS. Similarly, STATE_ART shows significant differences in Precision with p-values of 0.03882 when compared to various SCSA-PLUS versions, underscoring its distinct performance in this metric as well.

The results suggest statistically significant differences in MRR, MAP, NDCG, and Precision between B1, SCSA-PLUS, and between SCSA and SCSA-PLUS. Therefore, we reject the null hypothesis for all comparisons, suggesting that SCSA-PLUS significantly differs from both B1 and SCSA in terms of MRR, MAP, NDCG, and Precision.

5.7. Exploratory Analysis

This section presents two analyses to explore the tweets' features using a correlation matrix and why some tweets moved up or down in the ranking using the proposed algorithm.

5.7.1. Correlation Matrix of Recommended Tweets

Figure 21 shows the correlation matrix about how features are more correlated. The main idea is to answer questions about factors contributing to higher engagement. How does the sentiment of a tweet affect its diversity? How are the different metrics like retweets, likes, and quotes related? It is important to note that correlation does not imply causation. These correlations indicate relationships between variables but do not necessarily imply a direct cause-and-effect relationship.

- **Diversity and Sentiment:** The strong negative correlation (-0.71) between DiversityScore and SentimentScore is evident. This suggests that

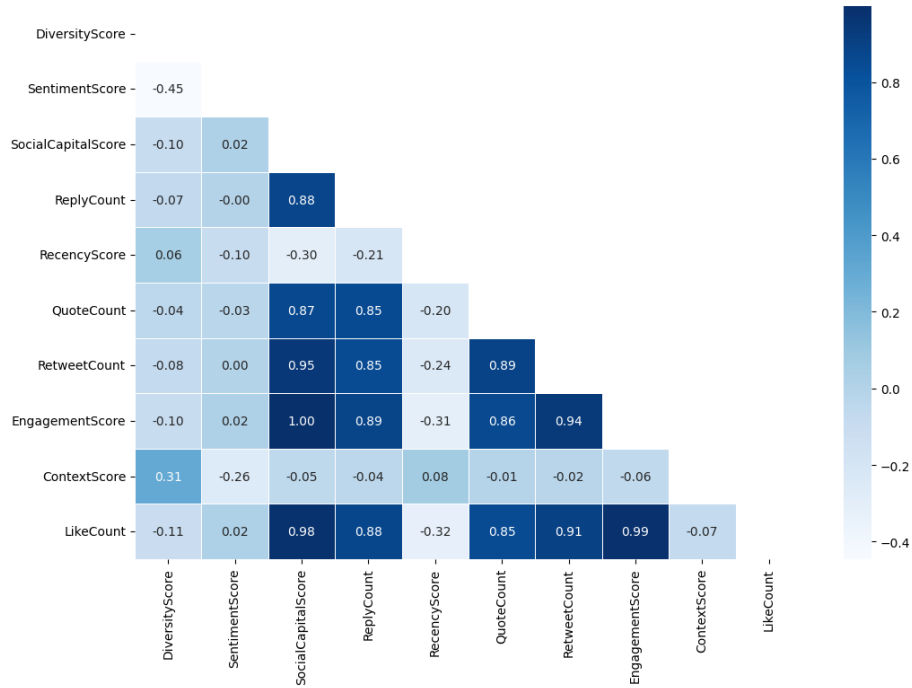


Figure 21: Correlation Matrix for Recommended Tweets.

tweets with a wider range of topics or ideas tend to have a more neutral or negative sentiment;

- **Engagement and Social Capital:** The strong positive correlation (0.99) between EngagementScore and SocialCapitalScore indicates that higher social capital is associated with higher engagement. This underscores the importance of building and maintaining social capital as a way to increase engagement with published content, for example;
- **Likes and Engagement:** The very strong positive correlation (0.99) between LikeCount and EngagementScore suggests that creating content that drives engagement will also result in a higher number of likes;
- **Retweets and Social Capital:** There is a strong positive correlation (0.89) between RetweetCount and SocialCapitalScore. This implies that

tweets from authors with higher social capital tend to get retweeted more often. This highlights the importance of building social capital to increase the reach of the content;

- **Quotes and Replies:** There is a strong positive correlation (0.73) between QuoteCount and ReplyCount. This suggests that tweets that generate more conversation (replies) are also more likely to be quoted. When creating content that aims to spark discussion or debate, this is important.

In summary, the correlation matrix provides valuable insights into the relationships between key variables in the dataset of Twitter data. These insights can be leveraged to understand news that drives engagement, likes, and conversation on the platform.

5.7.2. Analysis of Recommended Tweets

Considering each analyzed algorithm, we analyzed the recommended tweets generated by our proposal algorithm to identify what items moved up or down in the ranking list. In this case, we changed the ranking of items based on features of tweets and original ratings. The change in ranking was correlated with the engagement of the tweets, the social capital, and the number of likes, retweets, replies, and quotes of the tweets. The results indicate that the percentage of items that moved up or down in the ranking varies across different algorithms and original ratings. For example, for rating 1 of the algorithm SCSA-PLUS vs. B1, approximately 32.47% of the items moved up in the ranking, and 67.53% moved down. Similarly, for rating 5 of the algorithm SCSA-PLUS vs SCSA, approximately 56.28% of the items moved up in the ranking, and 43.72% moved down.

Table 4 presents the ranking difference was correlated with the other features in the dataset. The results indicate that the change in ranking is most strongly associated with the engagement of the tweets, the social capital, and the number of likes, retweets, replies, and quotes of the tweets.

The SentimentScore, RecencyScore, DiversityScore, and ContextScore have

Feature	Correlation with Ranking Difference
EngagementScore	0.178686
SocialCapitalScore	0.178487
LikeCount	0.176186
RetweetCount	0.153016
ReplyCount	0.133414
QuoteCount	0.122254

Table 4: Correlation of Features with Ranking Difference.

a lower correlation with the ranking difference, indicating that they have a lesser impact on the change in ranking, as shown in Table 5.

Feature	Correlation with Ranking Difference
SentimentScore	0.061083
RecencyScore	0.060732
DiversityScore	0.033455
ContextScore	0.029290

Table 5: Correlation of Features with Ranking Difference

The results suggest that items that receive higher engagement on social media are more likely to increase their ranking. This indicates that efforts to increase the engagement of tweets related to an item could potentially improve its ranking.

5.8. Discussion

The study reveals that the SCSA-PLUS algorithm demonstrates significant advancements over the B1 and SCSA algorithms across various metrics, including MRR, MAP@10, NDCG@10, and Precision at K. These improvements are substantiated by statistical significance and underscore the algorithm’s enhanced capability to generate relevant recommendations and effectively rank them according to user preferences.

While SCSA-PLUS outshines B1 in all metrics, confirming its efficiency and relevance, its performance relative to CS and SC suggests areas for refinement, particularly in ranking the most relevant items. Although it generally provides more accurate top 10 recommendations than CS, as indicated by a higher MAP@10, it trails slightly behind CS in MRR and NDCG@10. This implies that CS may better identify and rank the single most relevant item, even if its overall top recommendations are less accurate.

When compared with the STATE_ART algorithm, SCSA-PLUS shows promising results, particularly in MRR and MAP, suggesting it effectively identifies and ranks relevant items. However, the STATE_ART often outperforms SCSA-PLUS in NDCG, especially at higher cutoffs, highlighting its strength in placing the most relevant items at the top of the recommendations. This distinction is crucial, especially in contexts where the initial recommendations have a profound impact on user satisfaction and engagement.

The Precision@K analysis further supports the superior performance of SCSA-PLUS over B1 and SCSA, while also indicating a competitive edge against SC. However, both the CS algorithm and the STATE_ART demonstrate areas where they surpass SCSA-PLUS, particularly in the precision of ranking the most relevant items, suggesting a vital area for further enhancement.

The study also acknowledges the specific strengths of the Cosine Similarity (CS) algorithm. CS's direct approach to measuring content features can yield more precise recommendations for certain datasets or user preferences. In contrast, SCSA-PLUS, with its integration of social capital and sentiment analysis (SC+SA), provides a broader understanding of user preferences and social dynamics. This holistic approach is especially beneficial where social interactions and user sentiment significantly influence preferences.

In conclusion, while SCSA-PLUS shows notable advancements and potential, the comparison with STATE_ART highlights the need for further improvements, particularly in ranking the most relevant items at the top. Enhancing this aspect, possibly by integrating some strengths of the CS and STATE_ART algorithms, could make SCSA-PLUS more robust and effective. Future research might explore incorporating content-based features akin to CS and refining the ranking techniques to align with the precision demonstrated by STATE_ART, potentially leading to a more adaptable and powerful recommendation system.

5.9. Limitations

This study, while contributing significantly to the field of recommender systems, has certain limitations that warrant further discussion and consideration:

- **Lack of Online Evaluation:** Our research was constrained to offline evaluation, which might not fully reflect social media platforms' interactive and evolving nature. For example, an online evaluation could provide insights into how users engage with recommendations in real-time, revealing aspects like click-through rates and user dwell time that offline datasets cannot capture. Moving forward, integrating live user feedback and interaction data in an online setting would offer a more holistic understanding of the model's effectiveness;
- **Exclusion of Knowledge-Enhancement:** The model's deliberate omission of knowledge-enhancement, such as integrating external databases or text corpora, limits its ability to leverage a broader context in recommendations. For instance, incorporating semantic data or external content sources could enrich user profiles and recommendations, offering more nuanced and contextually relevant suggestions;
- **Absence of Community Formation Consideration:** The current model does not account for user community dynamics, which can significantly influence preferences and interactions on social platforms. For

instance, on platforms like Reddit or Twitter, user communities often form around specific topics or interests, influencing content virality and user engagement patterns. Future models could benefit from incorporating algorithms that detect and leverage these community structures to enhance recommendation relevance;

- **Model Complexity and Scalability:** Our model’s complexity poses scalability challenges, particularly when processing large-scale social media data. This limitation becomes apparent in scenarios requiring real-time data processing or handling diverse data types (e.g., text, images, videos). Future iterations of the model will need to focus on optimizing algorithms and data processing techniques to enhance scalability and efficiency, perhaps through methods like parallel processing or cloud-based architectures. We acknowledge its crucial role in ensuring the efficiency and effectiveness of our recommender system. Currently, our focus has been on establishing a solid foundation and ensuring the quality of our recommendations. We fully intend to address runtime and scalability in the upcoming stages of our project, once the core functionality has been validated and we can allocate the necessary resources;
- **Limited Evaluation Metrics:** The study primarily utilized conventional accuracy metrics, potentially overlooking aspects like recommendation diversity or user satisfaction. For instance, a system might have high accuracy but low diversity, leading to a "filter bubble" effect where users are only exposed to limited viewpoints. Integrating a wider range of metrics, including those that measure diversity, novelty, and user engagement, will provide a more comprehensive assessment of the system’s performance;
- **Single-Platform Focus:** Our research is based on data from Twitter, which may not fully represent user behaviors across various social media platforms. Different platforms (like Facebook, Instagram, or LinkedIn) have distinct user demographics, content types, and interaction patterns.

Future research could extend the model to other platforms, exploring how platform-specific dynamics affect recommendation strategies;

- **Empirical Analysis Against Baselines:** The current empirical analysis lacks a comprehensive comparison with state-of-the-art models. For a robust validation of our model’s performance, it would be beneficial to compare it against leading models in the field. This comparison would help identify the model’s unique strengths and areas for improvement in the context of the latest advancements in recommender systems;
- **Variability in Decay Schemes:** The study employed a specific logarithmic decay scheme for the recency score, which may not be optimal across different content types and user behaviors. Experimenting with various decay schemes, such as linear or exponential, could provide insights into their impact on content relevance over time. For example, news-related content might require a different decay approach than evergreen content, reflecting the varying temporal relevance of different content types. Although we varied the constant a between 0.1-1.0, more experiments need to be done with different data sets to find out which value is the most suitable.

Addressing these limitations in future research will refine the model and contribute valuable insights to the evolving field of recommender systems, particularly in adapting to the diverse and dynamic landscape of social media platforms.

6. Conclusion

This paper introduces significant innovations in the realm of recommender systems. The social capital model is a novel contribution that leverages user social connections and activities to enhance personalization, a feature not extensively explored in previous works. Additionally, the user model we developed,

deriving from user interactions and other characteristics, offers a unique approach to user profile construction. While user interaction profiles are indeed a common aspect in RSs, our approach uniquely integrates social capital aspects derived from social connections and activities, which needs to be extensively explored in existing models. We believe this integration offers a fresh perspective and advances the personalization capabilities of recommender systems

Finally, our offline evaluation method provides valuable insights that contribute to the advancement of recommender systems, specifically by integrating social network analysis and deepening understanding of social capital’s role in user behavior and preferences. In addition, this offline evaluation performs a comparison among SCSA-PLUS with other several literature methods to measure the quality of generated recommendations. The evaluation employed a comprehensive dataset obtained from previous work and involved a thorough analysis using various ranking metrics such as MRR, MAP, NDCG, and Precision.

The main contributions of this study are as follows:

- **SCSA-PLUS Algorithm:** A novel recommendation algorithm that significantly outperforms the baseline (B1) and its predecessor (SCSA) in terms of MRR, MAP, NDCG, and Precision, as supported by the statistical significance of the differences observed in the hypothesis tests. It also performs comparably to the SC algorithm, underscoring its effectiveness in generating relevant recommendations and ranking them in an order that aligns closely with user preferences.
- **Comprehensive Evaluation:** A thorough offline evaluation of the SCSA-PLUS algorithm using a comprehensive dataset and various ranking metrics. This evaluation provides valuable insights into the strengths and limitations of the SCSA-PLUS algorithm and serves as a benchmark for future studies in this area;
- **Insights into Recommendation Quality:** The study provides valuable insights into the relationships between key variables in the dataset of

Twitter data and how they affect the ranking of recommendations. For example, the results suggest that items that receive higher engagement on social media are more likely to increase the ranking. This indicates that efforts to increase the engagement of tweets related to an item could potentially improve its ranking.

For future work, our primary focus will be on conducting an online evaluation of our proposal to gain a more realistic assessment of its performance. This approach will allow us to closely examine how changes in the ranking of recommended tweets, influenced by factors such as engagement, social capital, and user interactions, impact overall effectiveness. We also plan to incorporate additional features like content analysis, network structure, and temporal dynamics to further enhance recommendation quality. Additionally, we will explore the use of alternative recommendation algorithms, including matrix factorization-based, deep learning-based, and hybrid algorithms, to potentially improve the system’s capability.

Another significant area of future development involves broadening our evaluation criteria. We aim to include a more diverse array of metrics, such as diversity, novelty, and user satisfaction, alongside traditional accuracy measures, to provide a more comprehensive evaluation of our recommender system’s performance. In addition, recognizing the potential benefits of knowledge enhancement, we will consider integrating external knowledge sources into our model. This integration aims to complement our current focus on the novel use of social capital, potentially enriching the recommender system with broader insights and contextual understanding.

Further, we intend to extend our empirical analysis to include additional state-of-the-art baselines, allowing for a more comprehensive comparative analysis and further validation of our model’s effectiveness. Alongside this, we will investigate the performance of our system across different decay schemes, particularly examining the effects of varying decay rates to enhance the model’s robustness.

Lastly, addressing the critical aspects of runtime and scalability is a key objective. We plan to conduct comparative tests under various load conditions to evaluate and ensure the efficiency and scalability of our model. These efforts will collectively contribute to refining our recommender system, ensuring it meets the dynamic demands of real-world applications.

References

- Abel, F., Gao, Q., Houben, G.-J., & Tao, K. (2013). Twitter-based user modeling for news recommendations. In *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence IJCAI '13* (p. 2962–2966). AAAI Press.
- Adomavicius, G., & Tuzhilin, A. (2005). Toward the next generation of recommender systems: a survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, 17, 734–749. URL: <https://doi.org/10.1109/tkde.2005.99>. doi:10.1109/tkde.2005.99.
- Adomavicius, G., & Tuzhilin, A. (2015). Context-aware recommender systems. In *Recommender Systems Handbook* (pp. 191–226). Springer US. URL: https://doi.org/10.1007/978-1-4899-7637-6_6. doi:10.1007/978-1-4899-7637-6_6.
- Bakshy, E., Rosenn, I., Marlow, C., & Adamic, L. (2012). The role of social networks in information diffusion. URL: <https://arxiv.org/abs/1201.4145>. doi:10.48550/ARXIV.1201.4145.
- Bobadilla, J., Ortega, F., Hernando, A., & Gutiérrez, A. (2013). Recommender systems survey. *Knowledge-Based Systems*, 46, 109–132. URL: <https://doi.org/10.1016/j.knosys.2013.03.012>. doi:10.1016/j.knosys.2013.03.012.

- Bourdieu, P. (2002). The forms of capital. In *Readings in Economic Sociology* (pp. 280–291). Blackwell Publishers Ltd. doi:10.1002/9780470755679.ch15.
- Cañamares, R., Castells, P., & Moffat, A. (2020). Offline evaluation options for recommender systems. *Information Retrieval Journal*, 23, 387–410. URL: <https://doi.org/10.1007/s10791-020-09371-3>. doi:10.1007/s10791-020-09371-3.
- Chen, J., Xin, X., Liang, X., He, X., & Liu, J. (2022). GDSRec: Graph-based Decentralized Collaborative filtering for Social Recommendation. *IEEE Transactions on Knowledge and Data Engineering*, (pp. 1–11). URL: <https://doi.org/10.1109/tkde.2022.3153284>. doi:10.1109/tkde.2022.3153284.
- Coleman, J. (1990). *Foundations of social theory*. Cambridge, Mass: Belknap Press of Harvard University Press.
- Cui, P., Yin, B., & Xu, B. (2023). The application of social recommendation algorithm integrating attention model in movie recommendation. *Scientific Reports*, 13. URL: <http://dx.doi.org/10.1038/s41598-023-43511-1>. doi:10.1038/s41598-023-43511-1.
- Ge, M., Persia, F., & D'Auria, D. (2019). Advanced recommender systems by exploiting social networks. In *2019 IEEE International Conference on Humanized Computing and Communication (HCC)*. IEEE. URL: <https://doi.org/10.1109/hcc46620.2019.00025>. doi:10.1109/hcc46620.2019.00025.
- Girgin, B. A. (2021). Ranking influencers of social networks by semantic kernels and sentiment information. *Expert Systems with Applications*, 171, 114599. URL: <https://doi.org/10.1016/j.eswa.2021.114599>. doi:10.1016/j.eswa.2021.114599.

- Gonzalez-Camacho, L. A., Faria, J. H. K., Machado, L. T., & Alves-Souza, S. N. (2022). Recommender system based on the friendship between social network users in a cold-start scenario. In *Information Systems and Technologies* (pp. 234–252). Springer International Publishing. URL: https://doi.org/10.1007/978-3-031-04829-6_21. doi:10.1007/978-3-031-04829-6_21.
- Guo, Z., Cho, J.-H., Chen, I.-R., Sengupta, S., Hong, M., & Mitra, T. (2022). SAFER: Social capital-based friend recommendation to defend against phishing attacks. *Proceedings of the International AAAI Conference on Web and Social Media*, 16, 241–252. URL: <https://doi.org/10.1609/icwsm.v16i1.19288>. doi:10.1609/icwsm.v16i1.19288.
- He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T.-S. (2017). Neural collaborative filtering. URL: <https://arxiv.org/abs/1708.05031>. doi:10.48550/ARXIV.1708.05031.
- Horne, B. D., & Adali, S. (2017). This just in: Fake news packs a lot in title, uses simpler, repetitive content in text body, more similar to satire than real news. URL: <https://arxiv.org/abs/1703.09398>. doi:10.48550/ARXIV.1703.09398.
- Isinkaye, F., Folaajimi, Y., & Ojokoh, B. (2015). Recommendation systems: Principles, methods and evaluation. *Egyptian Informatics Journal*, 16, 261–273. doi:<https://doi.org/10.1016/j.eij.2015.06.005>.
- Kapoor, K. K., Tamilmani, K., Rana, N. P., Patil, P., Dwivedi, Y. K., & Nerur, S. (2017). Advances in social media research: Past, present and future. *Information Systems Frontiers*, 20, 531–558. doi:10.1007/s10796-017-9810-y.
- Karidi, D. P., Stavarakas, Y., & Vassiliou, Y. (2017). Tweet and followee personalized recommendations based on knowledge graphs. *Journal of Ambient Intelligence and Humanized Computing*, 9, 2035–2049.

- Lee, E., Kim, Y. J., & Ahn, J. (2014). How do people use facebook features to manage social capital? *Computers in Human Behavior*, *36*, 440–445. doi:10.1016/j.chb.2014.04.007.
- Li, C., & Xiong, F. (2017). Social recommendation with multiple influence from direct user interactions. *IEEE Access*, *5*, 16288–16296.
- Lin, N. (2001). *Social Capital: A Theory of Social Structure and Action*. Structural Analysis in the Social Sciences. Cambridge University Press. doi:10.1017/CB09780511815447.
- Lunardi, G. M., Machado, G. M., Maran, V., & de Oliveira, J. P. M. (2020). A metric for filter bubble measurement in recommender algorithms considering the news domain. *Applied Soft Computing*, *97*, 106771. URL: <https://doi.org/10.1016/j.asoc.2020.106771>. doi:10.1016/j.asoc.2020.106771.
- Morone, F., & Makse, H. A. (2015). Influence maximization in complex networks through optimal percolation, . URL: <https://arxiv.org/abs/1506.08326>. doi:10.48550/ARXIV.1506.08326.
- Nahapiet, J., & Ghoshal, S. (1998). Social capital, intellectual capital, and the organizational advantage. *Academy of Management Review*, *23*, 242–266. doi:10.5465/amr.1998.533225.
- Putnam, R. D. (2000). Bowling alone. In *Proceedings of the 2000 ACM conference on Computer supported cooperative work - CSCW '00*. ACM Press.
- Rezaie, B., Zahedi, M., & Mashayekhi, H. (2020). Measuring time-sensitive user influence in twitter. *Knowledge and Information Systems*, *62*, 3481–3508. URL: <https://doi.org/10.1007/s10115-020-01459-y>. doi:10.1007/s10115-020-01459-y.
- Ricci, F., Rokach, L., & Shapira, B. (2015). Recommender systems: Introduction and challenges. In F. Ricci, L. Rokach, & B. Shapira (Eds.), *Recommender Systems Handbook* chapter 1. (pp. 1–34). Springer US.

- Singh, D. K. S., N., N., L., R., Sanghavi, P., Vaghela, R. S., Manoharan, P., Hamdi, M., & Tunze, G. B. (2022). Social network analysis for precise friend suggestion for twitter by associating multiple networks using ML. *International Journal of Information Technology and Web Engineering*, 17, 1–11. URL: <https://doi.org/10.4018/ijitwe.304050>. doi:10.4018/ijitwe.304050.
- Spratt, C., Hong, J., McAreavey, K., & Liu, W. (2018). Community-based measures for social capital. In *Studies in Computational Intelligence* (pp. 327–338). Springer International Publishing. doi:10.1007/978-3-030-05414-4_26.
- Tiwari, S., Kumar, S., Jethwani, V., Kumar, D., & Dadhich, V. (2021). Pntrs: Personalized news and tweet recommendation system. *Journal of Cases on Information Technology*, 24, 1–19. URL: <http://dx.doi.org/10.4018/JCIT.20220701.oa9>. doi:10.4018/jcit.20220701.oa9.
- Tsai, C., Huhtamäki, J., Olsson, T., & Brusilovsky, P. (2020). Diversity exposure in social recommender systems: A social capital theory perspective. *CEUR Workshop Proceedings*, 2682, 57–64. Publisher Copyright: Copyright © 2020 for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).; 7th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems, IntRS 2020 ; Conference date: 26-09-2020.
- Zhang, R., Bao, H., Sun, H., Wang, Y., & Liu, X. (2015). Recommender systems based on ranking performance optimization. *Frontiers of Computer Science*, 10, 270–280. doi:10.1007/s11704-015-4584-1.
- Zhang, X. S., Zhang, X., & Kaparathi, P. (2020). *IEEE Transactions on Computational Social Systems*, 7, 924–939. doi:10.1109/TCSS.2020.3001093.
- Zhuang, Y.-B., Li, Z.-H., & Zhuang, Y.-J. (2021). Identification of influencers in online social networks: measuring influence considering multidimensional

factors exploration. *Heliyon*, 7, e06472. URL: <https://doi.org/10.1016/j.heliyon.2021.e06472>. doi:10.1016/j.heliyon.2021.e06472.